Radar Design Study
Part-B Trades Examples
Rev-B

EEE 5557 Intro to Radar

Design Trades Examples

The following design trades are presented in an arbitrary order – NOT optimized order. You should think about your trades before starting.

Example-1: UAV Operating Altitude Trade

- How does the selected Altitude affect other radar parameters?

For a selected design parameter: Does Alt impact?	*Alt* impact?

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Ant El HPBW 2-way: (L = footprint along-track = *f(Alt)* ?) (PRF for single pulse on surface? (Rfar – Rnear) = *f(Alt)*)	Yes Yes

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Ant El HPBW 2-way: (L = footprint along-track?) (PRF for single pulse on surface?) (Rfar – Rnear) = *f(Alt)*)	Yes Yes
Ant Az HPBW 2-way: (pulse IFOV, del-Az = *f(Alt)*?) (TOT – for fixed Antenna spin rate? TOT = f(Az HPBW/360°))	Yes No

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Ant gain/size: (for selected Ant El & Az HPBW?)	No

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Xmt pulse width (pulse IFOV del-R?)	No

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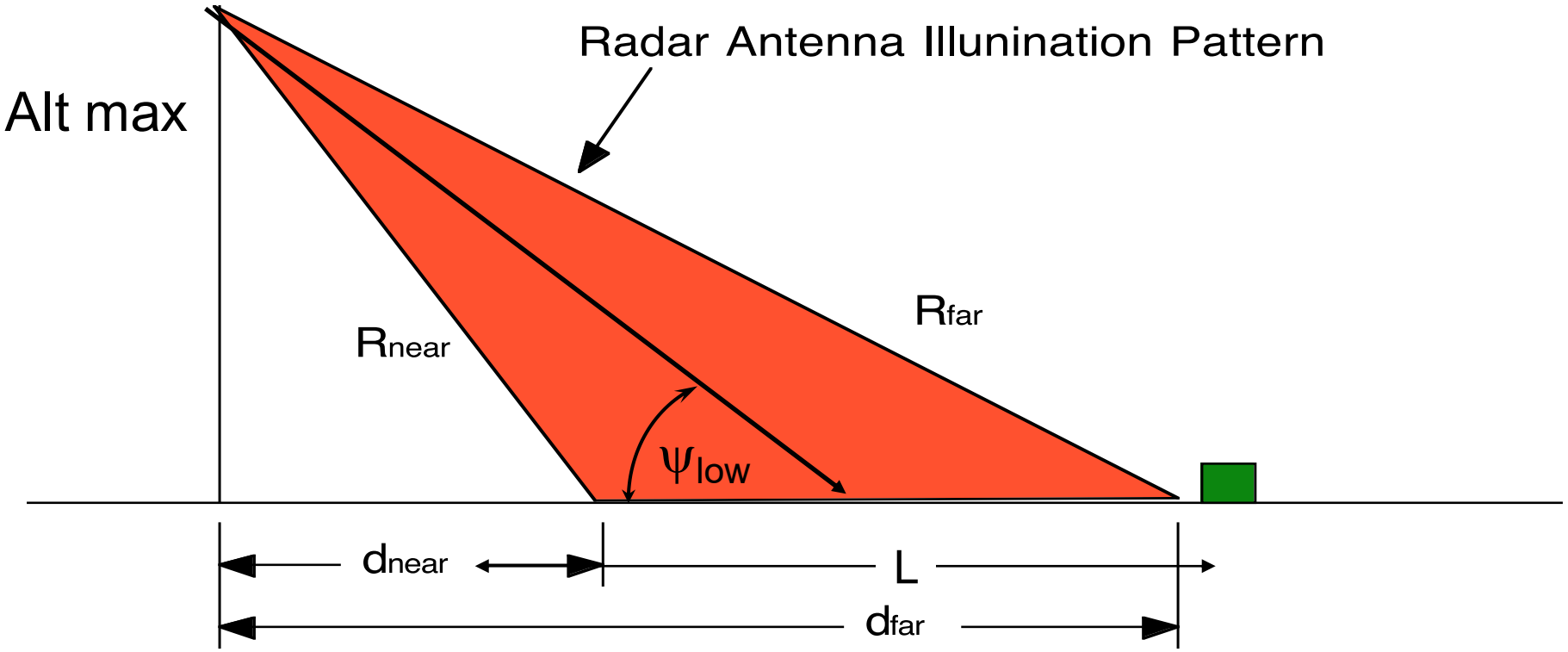
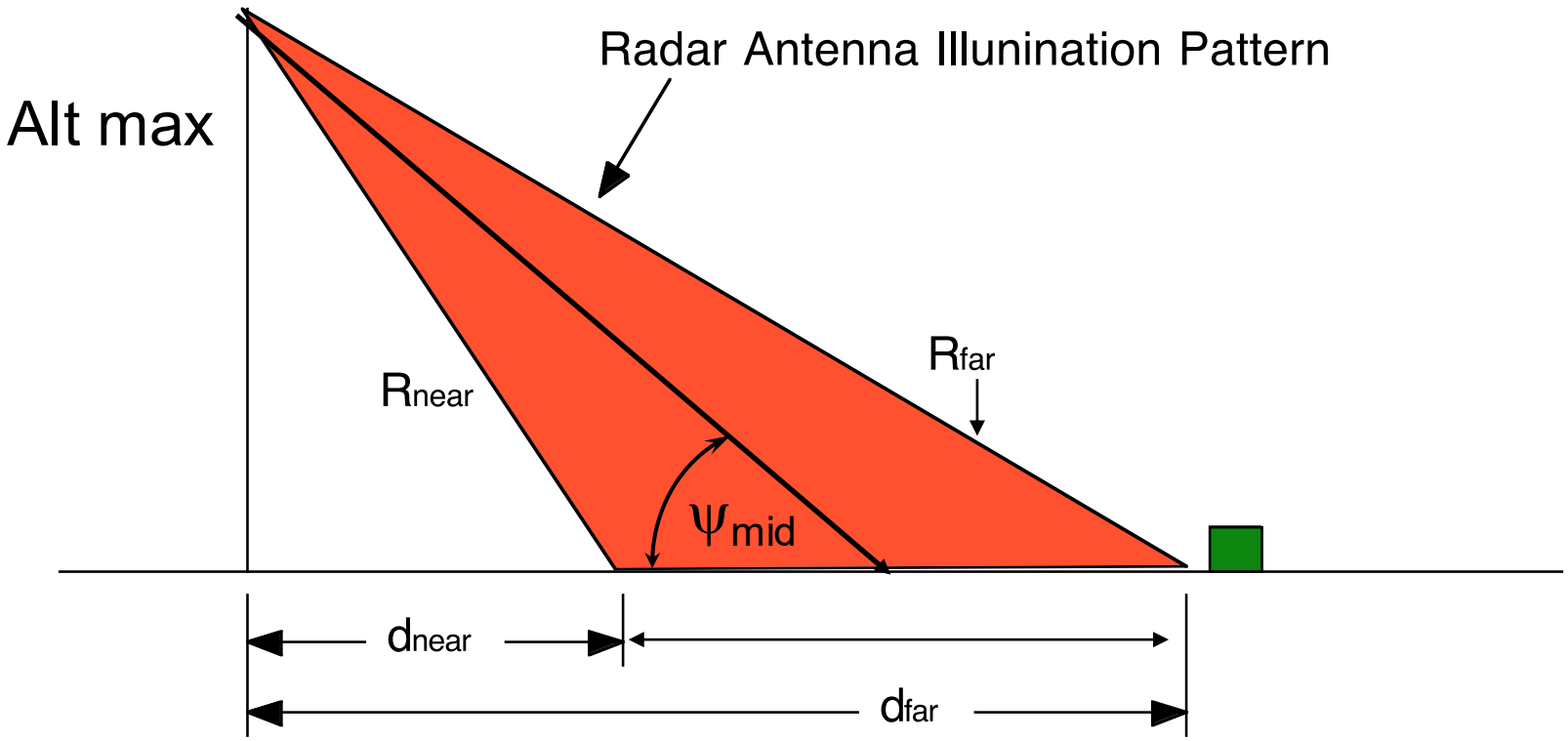
Ant gain/size: (for selected Ant El & Az HPBW?)	No
Ant boresight grazing ang?	No
Xmt pulse width (pulse IFOV del-R?)	No
UAV/Target velocity (PRF Doppler measurement?)	No

Example-1: UAV Altitude & Boresight Pointing Trade

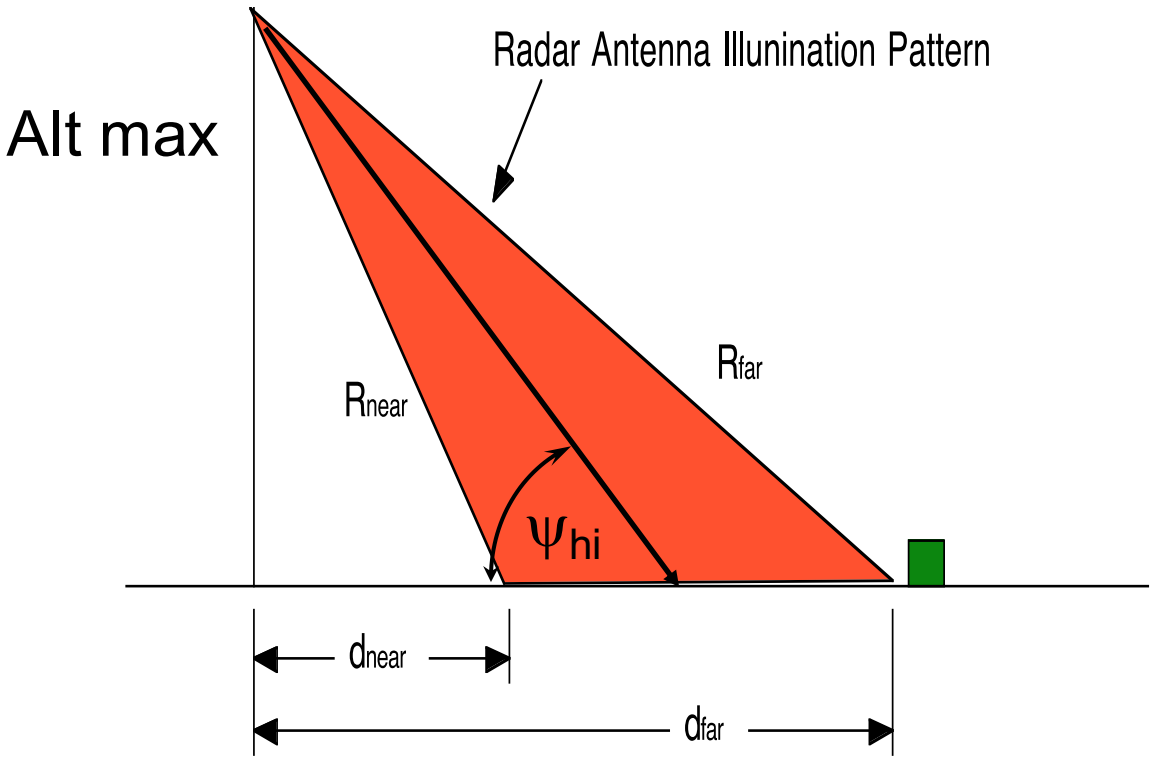
Find SNR as function of Grazing Ang



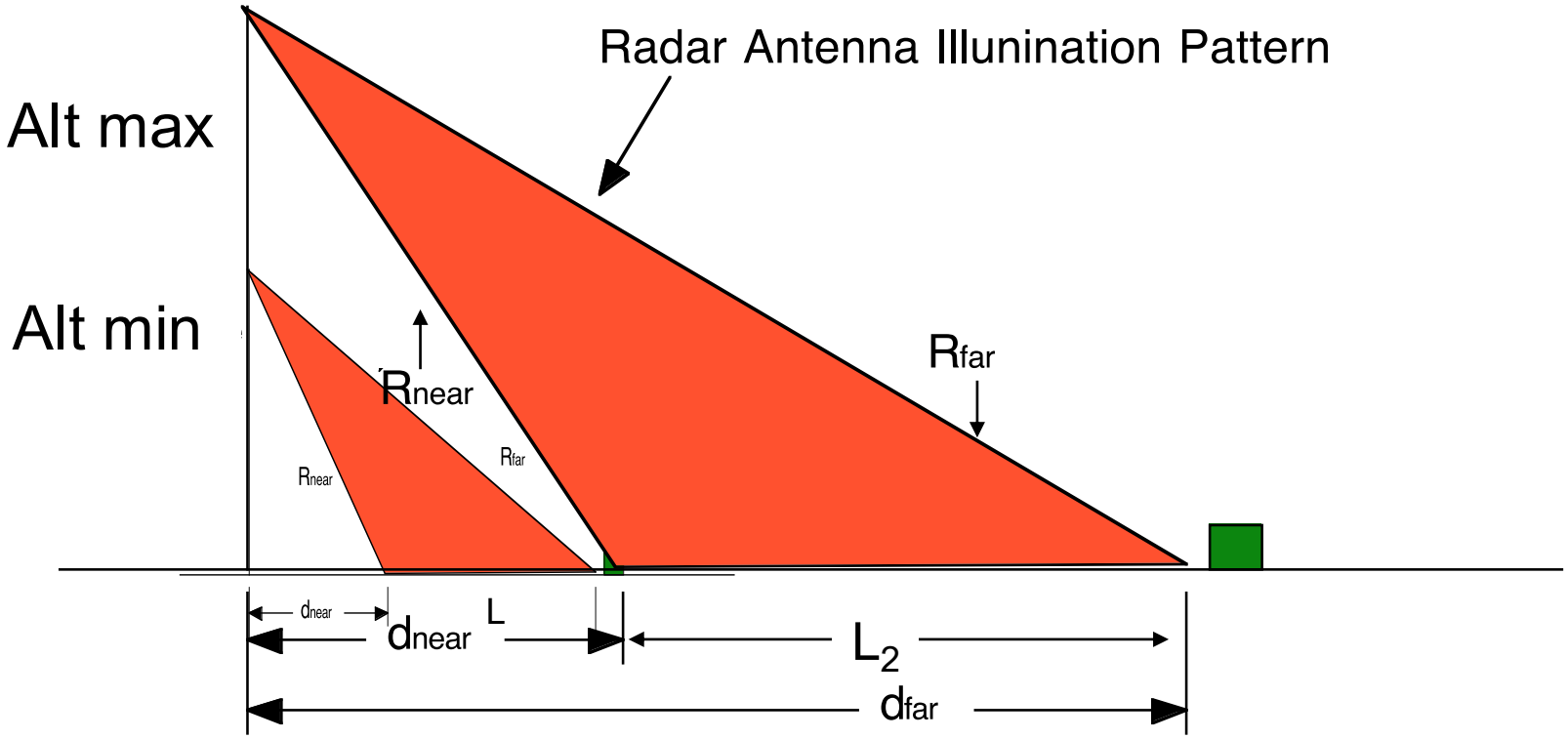
Find SNR as function of Grazing Ang



Find SNR as function of Grazing Ang



Find SNR as function of UAV Altitude



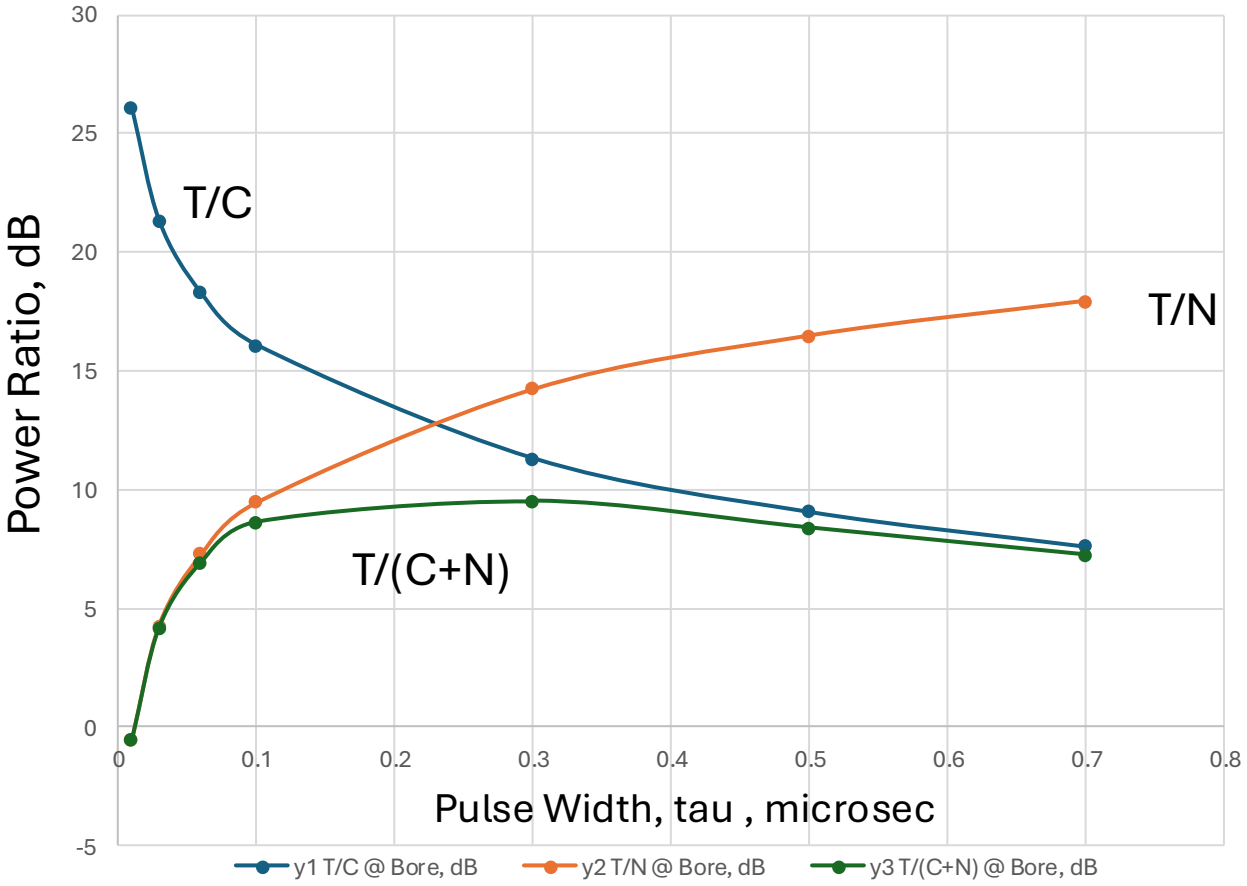
Examples of Trades for S26

- This presentation gives examples of typical results that you should obtain during your Radar design project
- They are meant to be examples that you should obtain, but NOT a recommended design
- DrJ has conducted numerous trade studies that indicate that there are many spec-compliant designs possible for many combinations of frequencies, altitudes and boresight pointing angles
  - Once you have a spec compliant design, you should optimize the design to meet (TBD) selected goals
    - All goals are equally weighted
    - It is highly unlikely that all goals can be simultaneously satisfied
  - Your Radar Design Proposal should present several (3 or more) spec-compliant designs and your rational for your final design selection

Signal to Noise Expectations
(with varying Pulse Width, tau)

All Constant

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant_Bore_grazing_Ang		
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmt power		W
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer



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Xmt power		W
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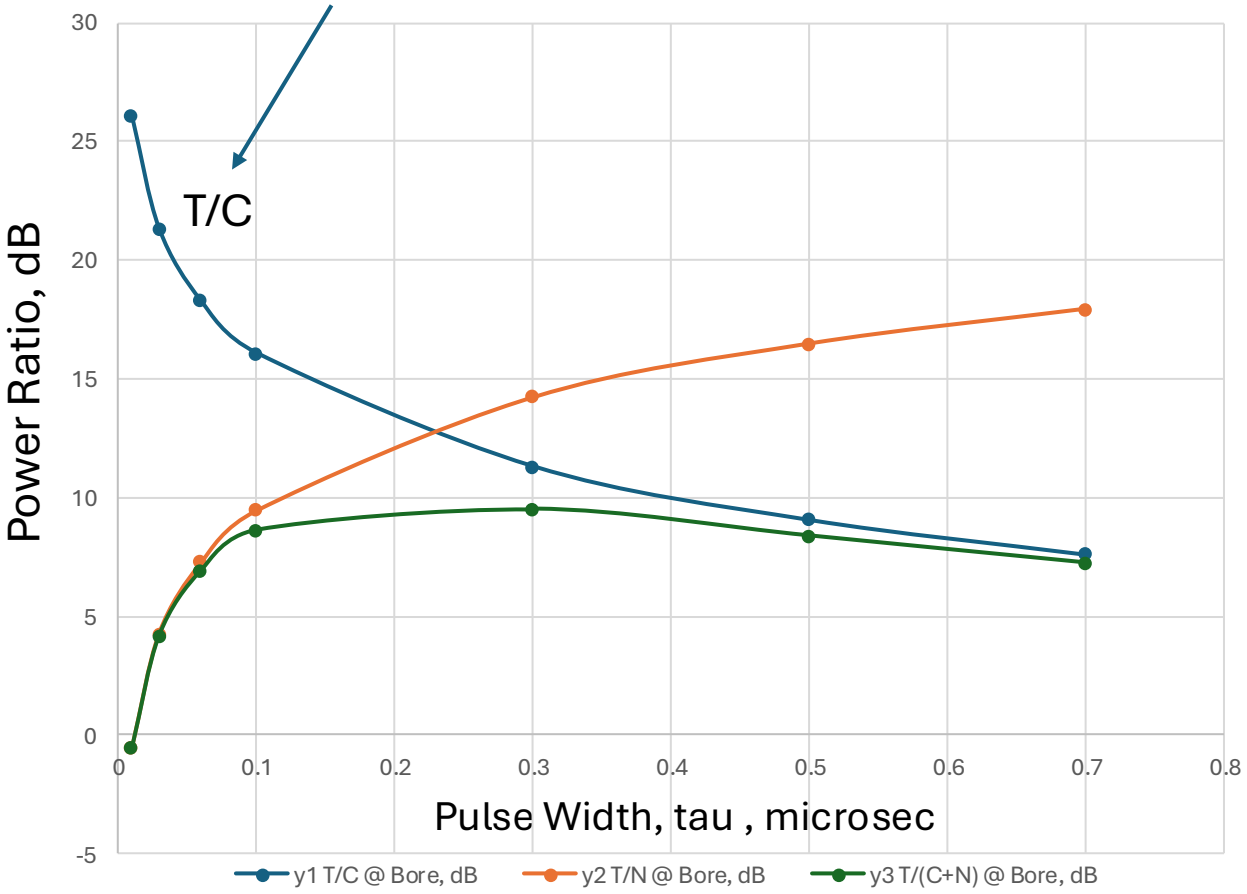
Design capability: # scans on target = N		integer

As Tau increases, pulse IFOV (δR) increases & Clutter increases



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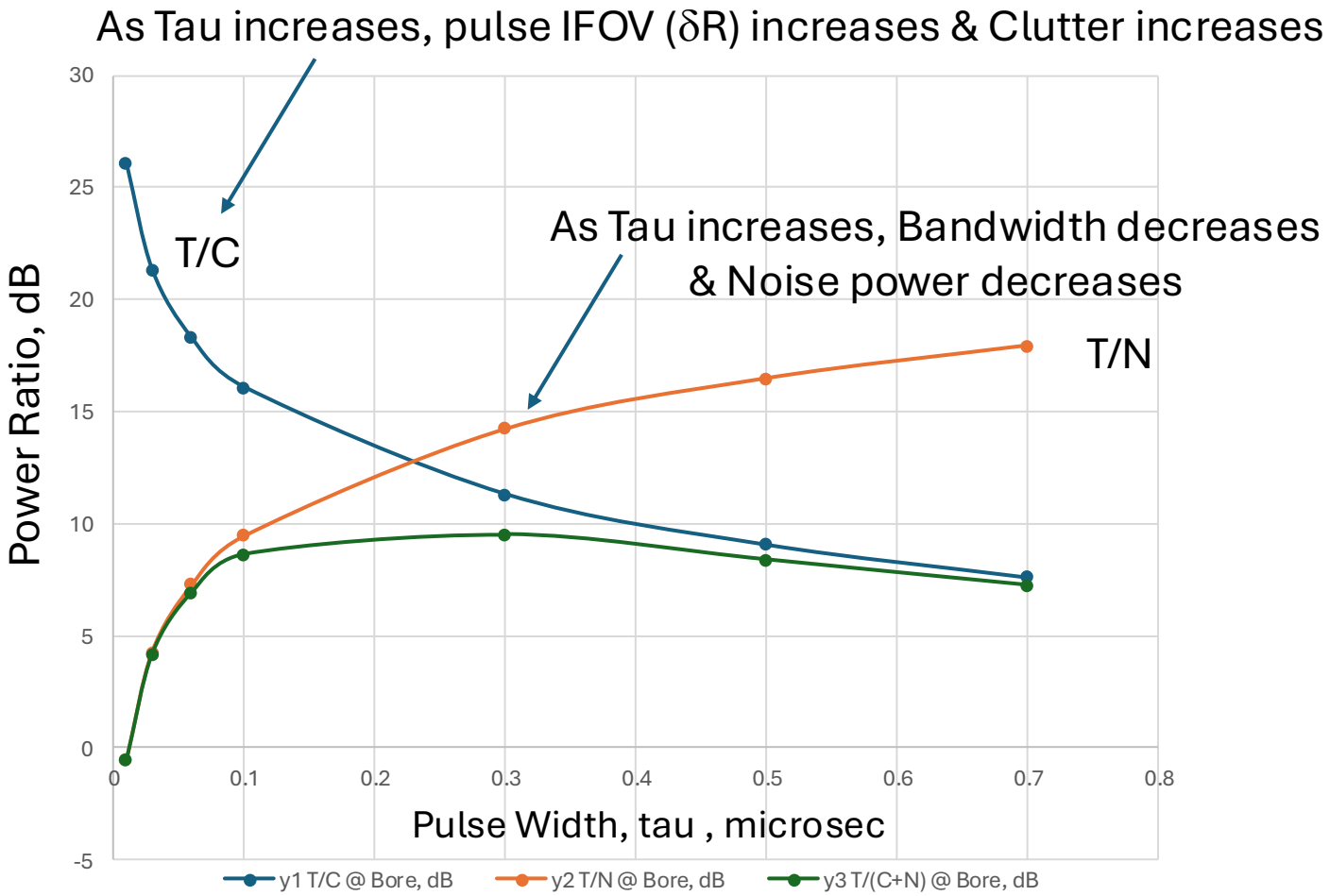

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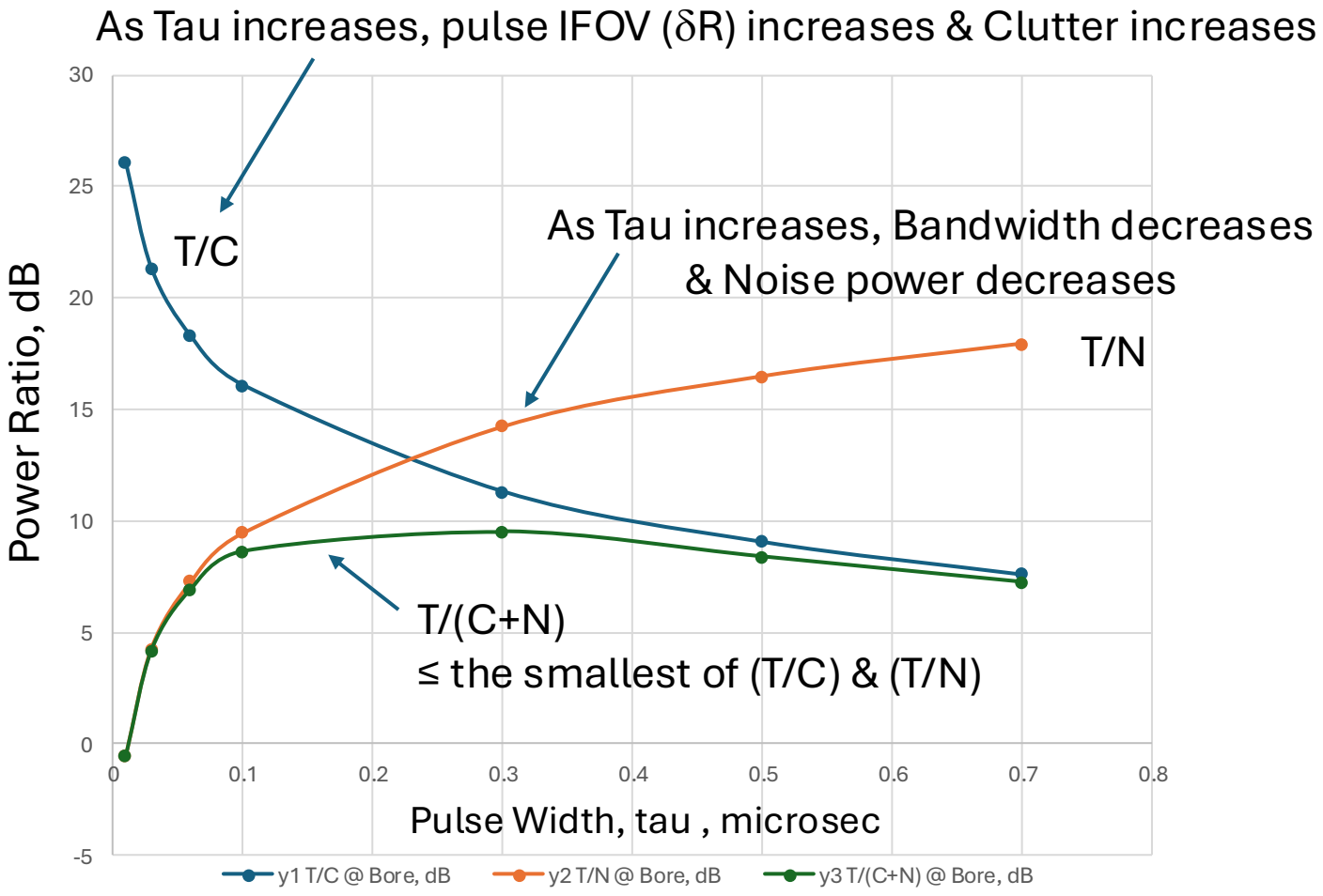
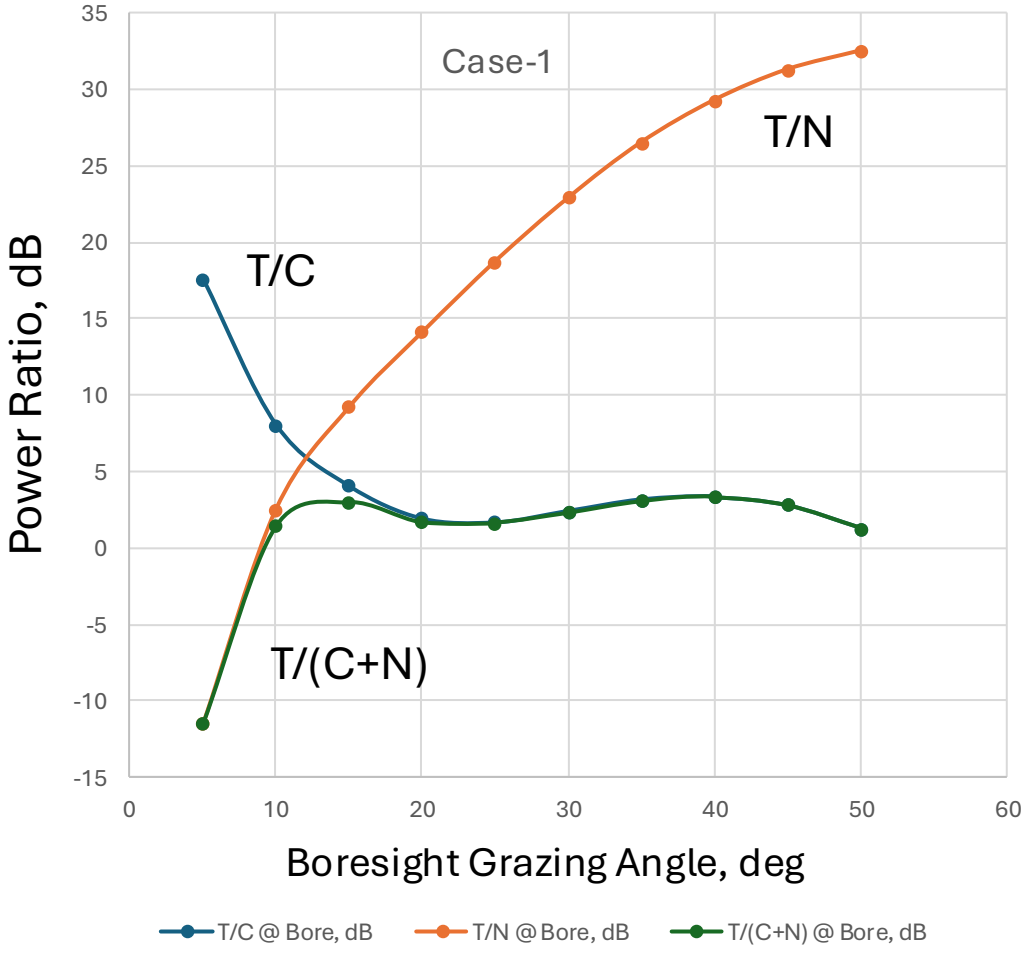


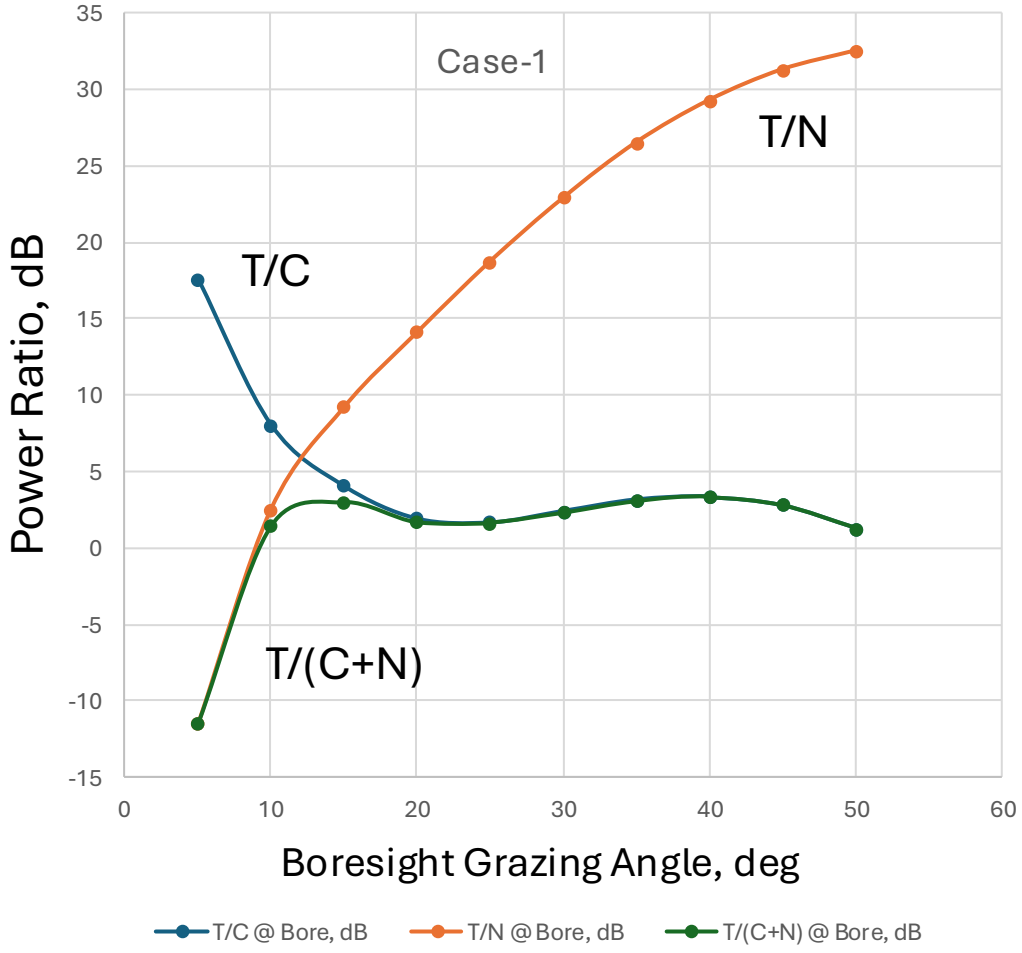
Signal to Noise Expectations

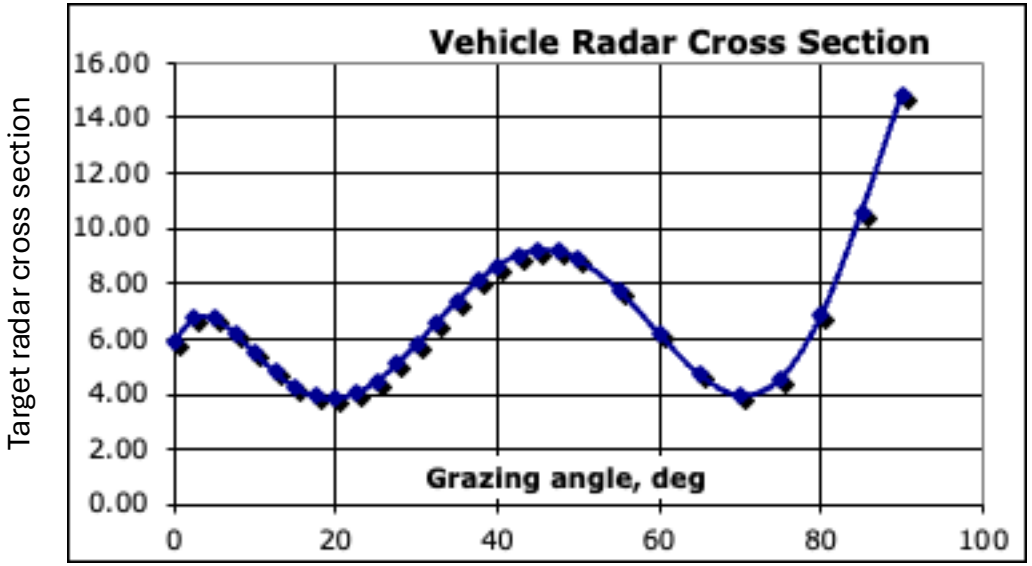
(with varying Boresight Grazing Angle)

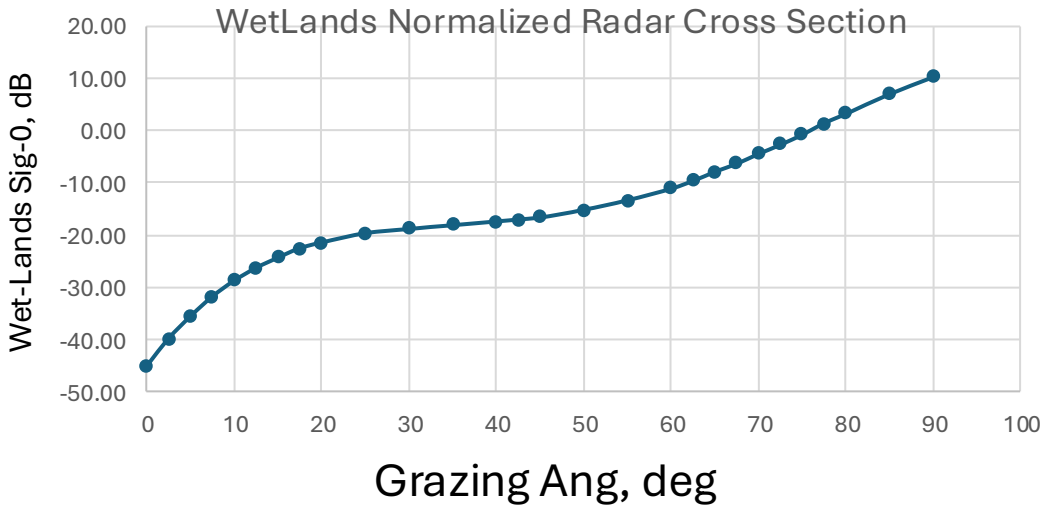


Signal to Noise Expectations

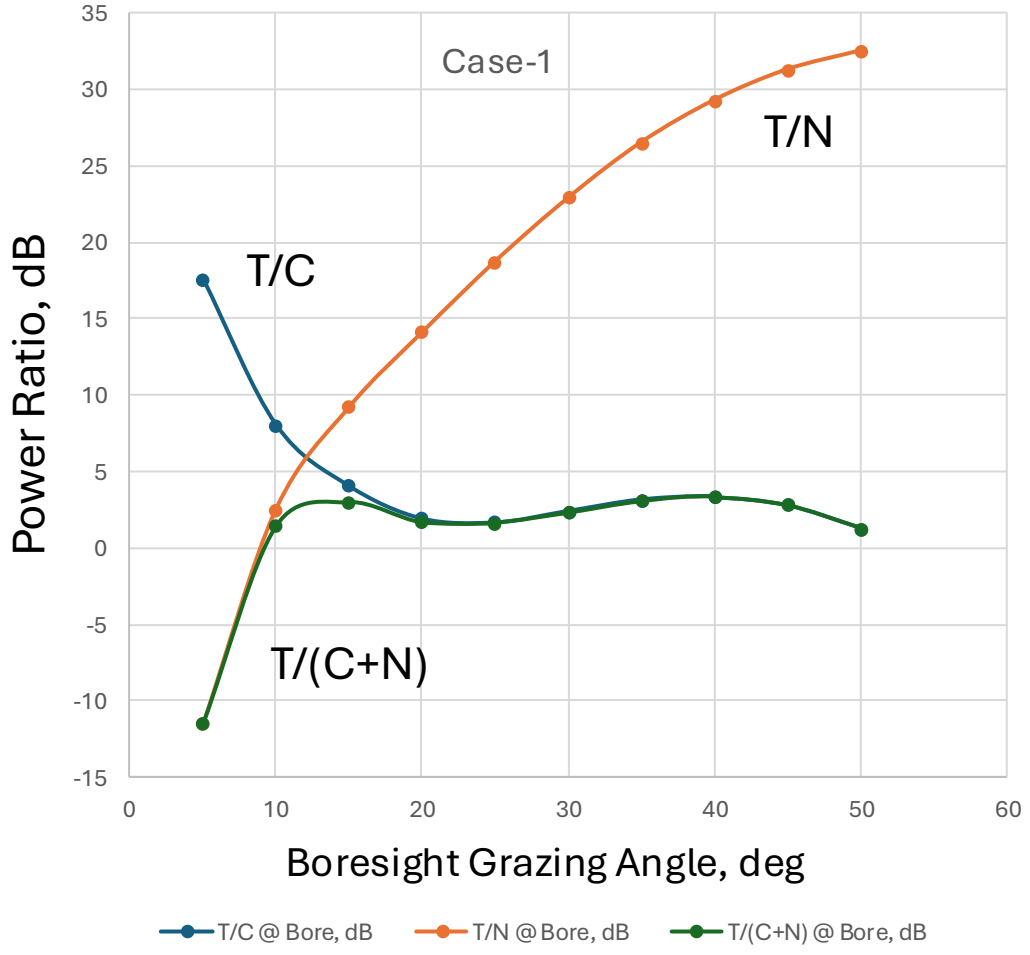
Shape of Curve Determined by Changes in Target and Clutter with Grazing Angle

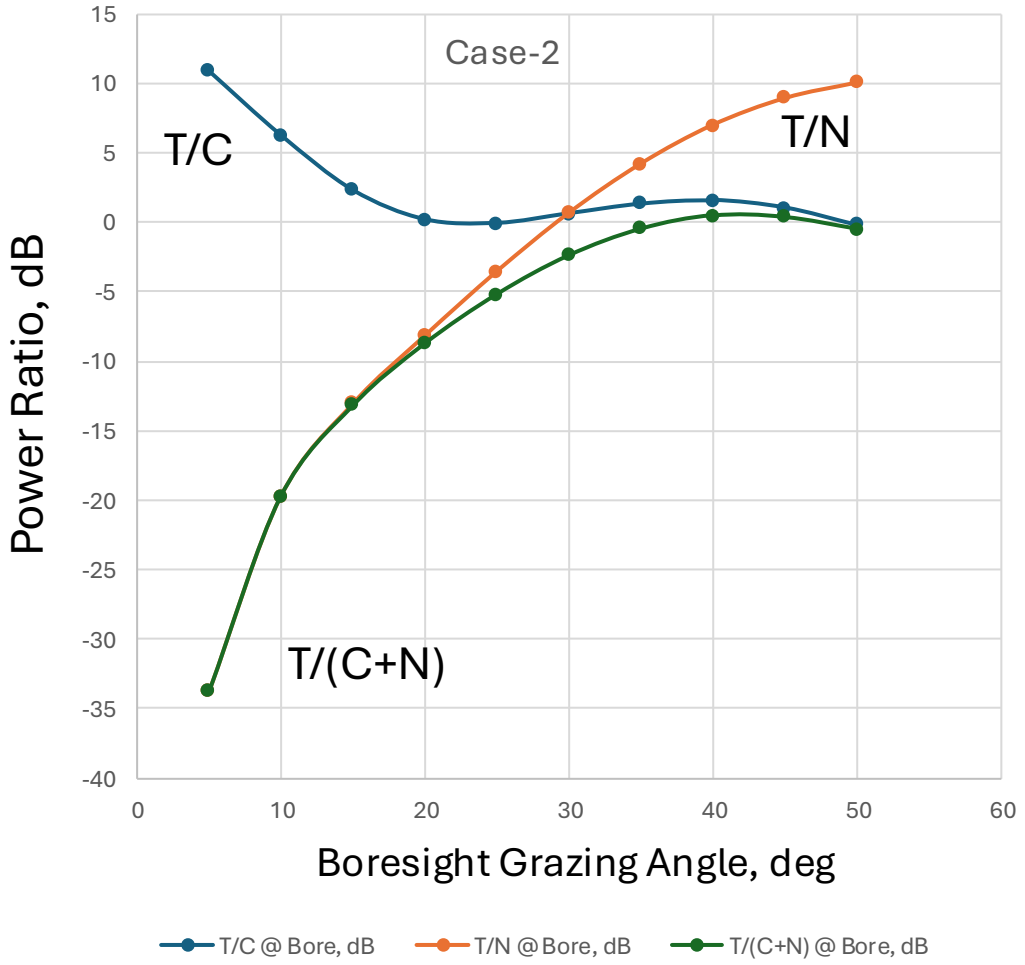




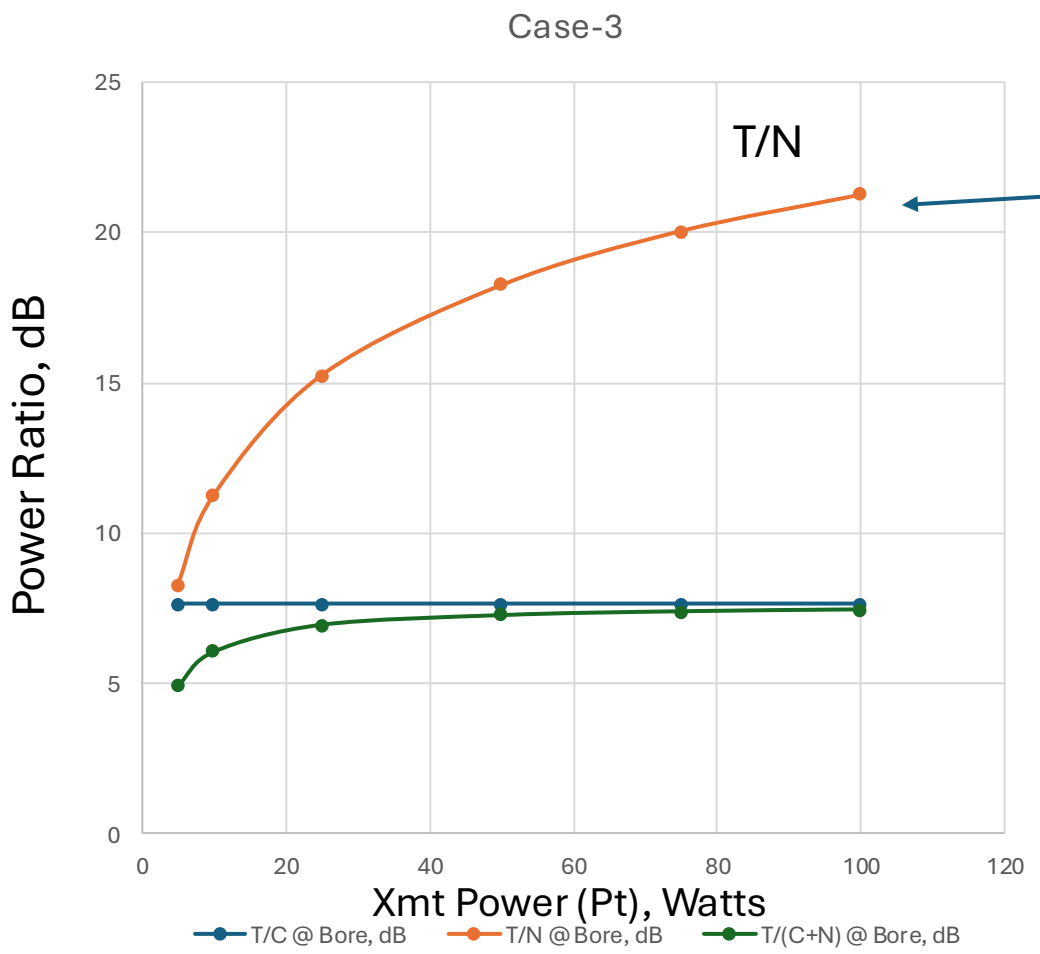


Signal to Noise Expectations
(with varying Boresight Grazing Angle)



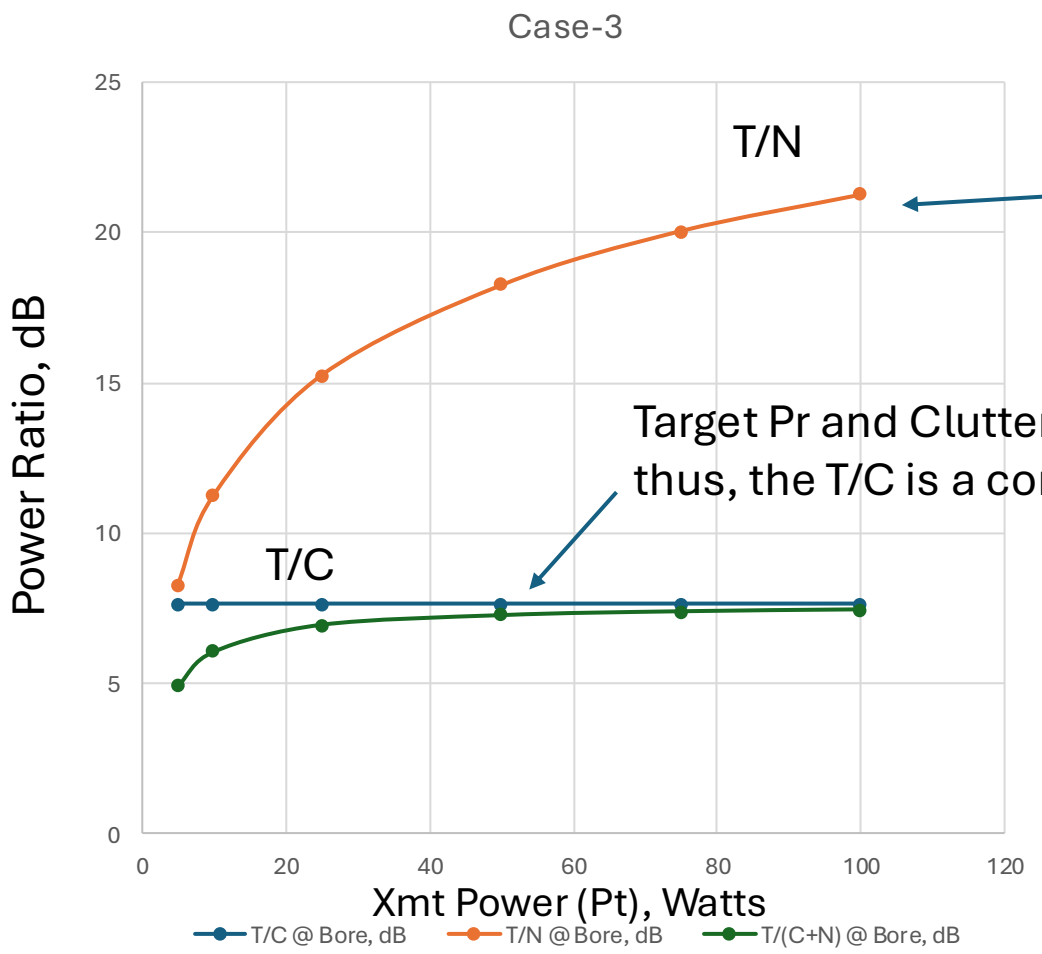


Signal to Noise Expectations (with varying Xmt Power, Pt)



Radar X-factor increases with Xmt Pwr (Pt); thus, Target Pr increases with Pt

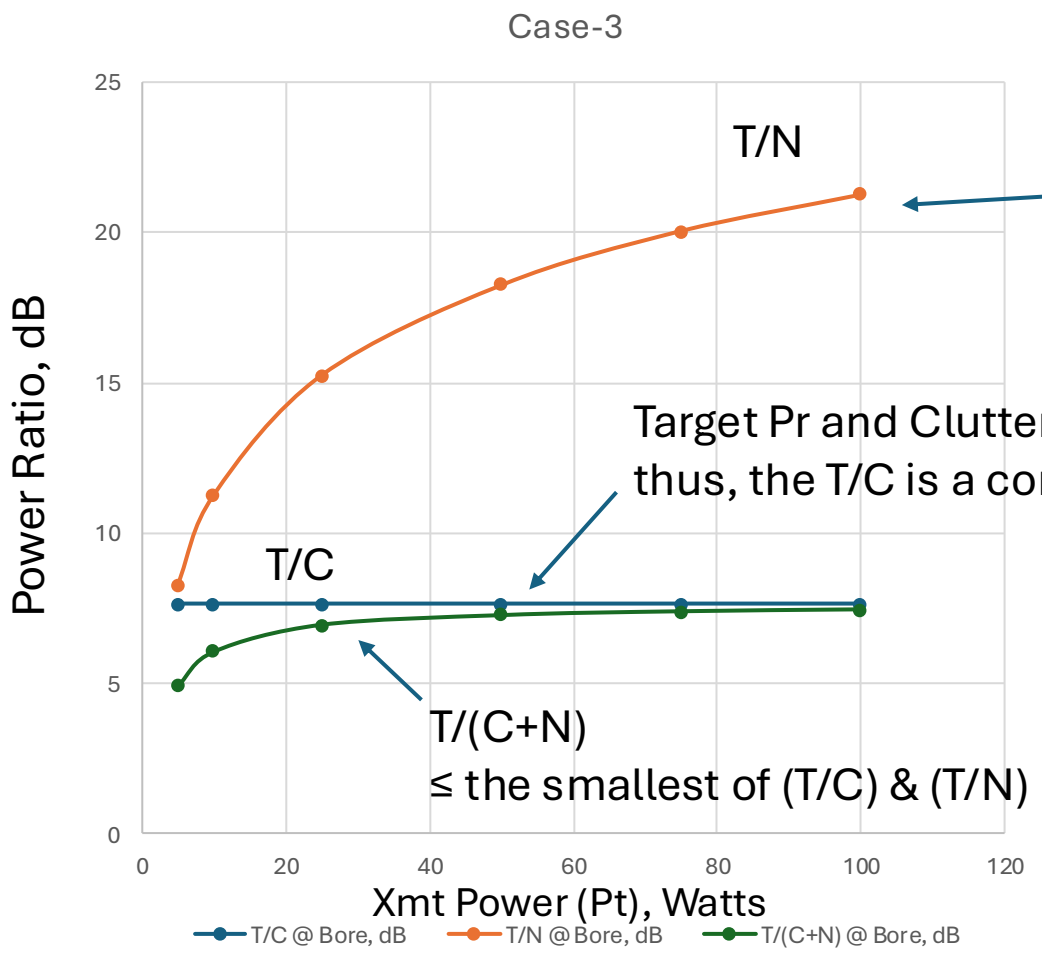
Signal to Noise Expectations (with varying Xmt Power, Pt)



Radar X-factor increases with Xmt Pwr (Pt); thus, Target Pr increases with Pt

Target Pr and Clutter Pr share a constant X-factor; thus, the T/C is a constant

Signal to Noise Expectations (with varying Xmt Power, Pt)



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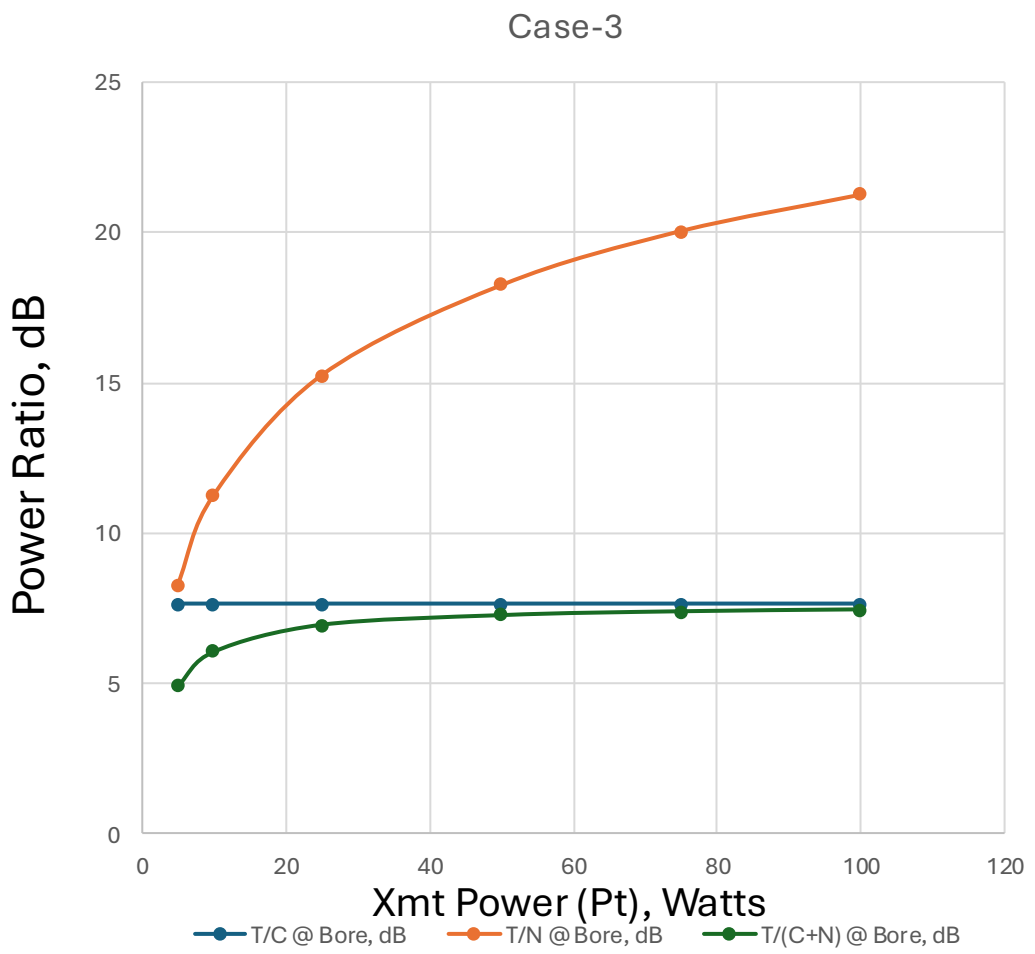

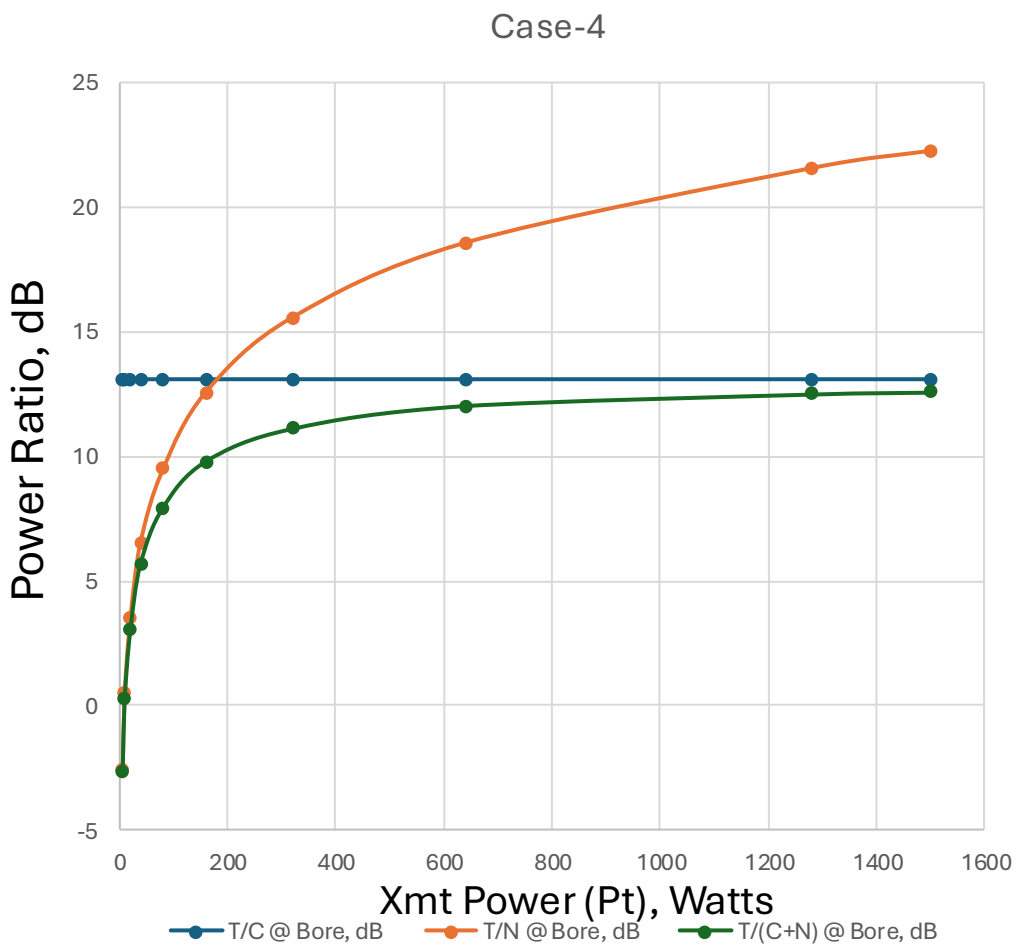

Example-2 Min Warning Time Req’mt Design Trades

Example Min Warning Time Reqmt Design Trades

- Given a min warning time req'mt
  - **Minimum Target Detection Time (warning time) T_{war}:** Using the worst-case target velocity (speed and direction of travel) and swath location, the **requirement** is:
    - to detect the target **at least 6 sec** before the aircraft reaches the target (i.e., the point of closest approach is when the target crosses the cross-track plane)
    - This requirement allows time for a camera with cross-track

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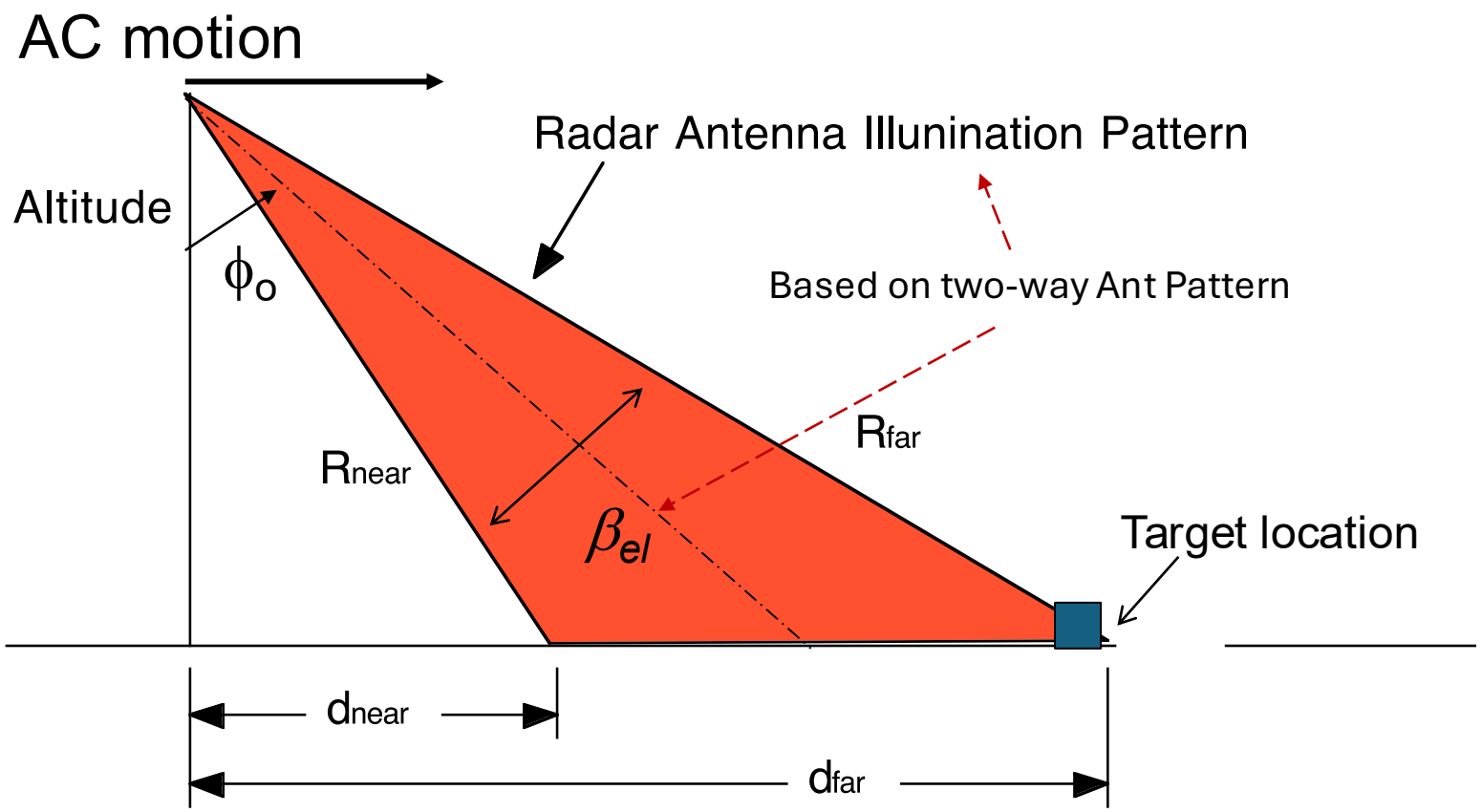
    - This requirement allows time for a camera with cross-track pointing capability to be pointed at the target as the aircraft flies overhead (or passes by)
    - The **goal** is to increase warning time up to **60 sec**. Bonus evaluation points will be assigned for this enhanced capability

Min Warning Time Reqmt Design Trades

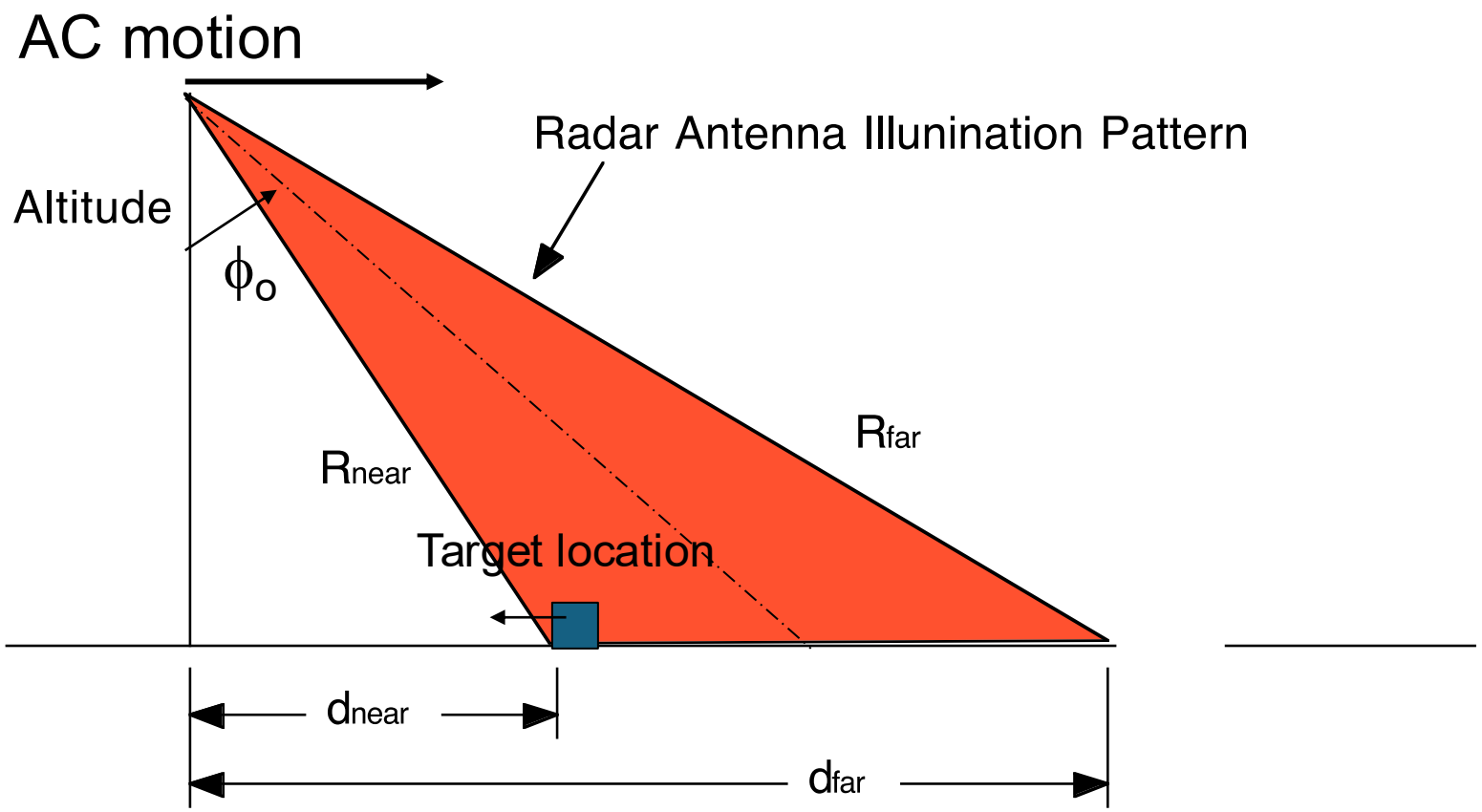
- Given a min warning time reqmt
  - **Minimum Target Detection Time (warning time) T_{war}:** Using the <u>worst-case target velocity (speed and direction of travel) and swath location</u>, the **requirement** is:
    - to detect the target **at least 3 sec** before the aircraft reaches the target (the point of closest approach is when the target crosses the cross-track plane)
    - This requirement allows time for a camera with cross-track pointing capability to be pointed at the target as the aircraft flies overhead (or passes by)
    - The **goal** is to increase warning time up to **60 sec**. Bonus evaluation points will be assigned for this enhanced capability
  - What parameters affect the T_{war}?

Antenna Footprint & Target Location for Best Case (Max Warning Time),

(Az = 0°and Target directly ahead of UAV)

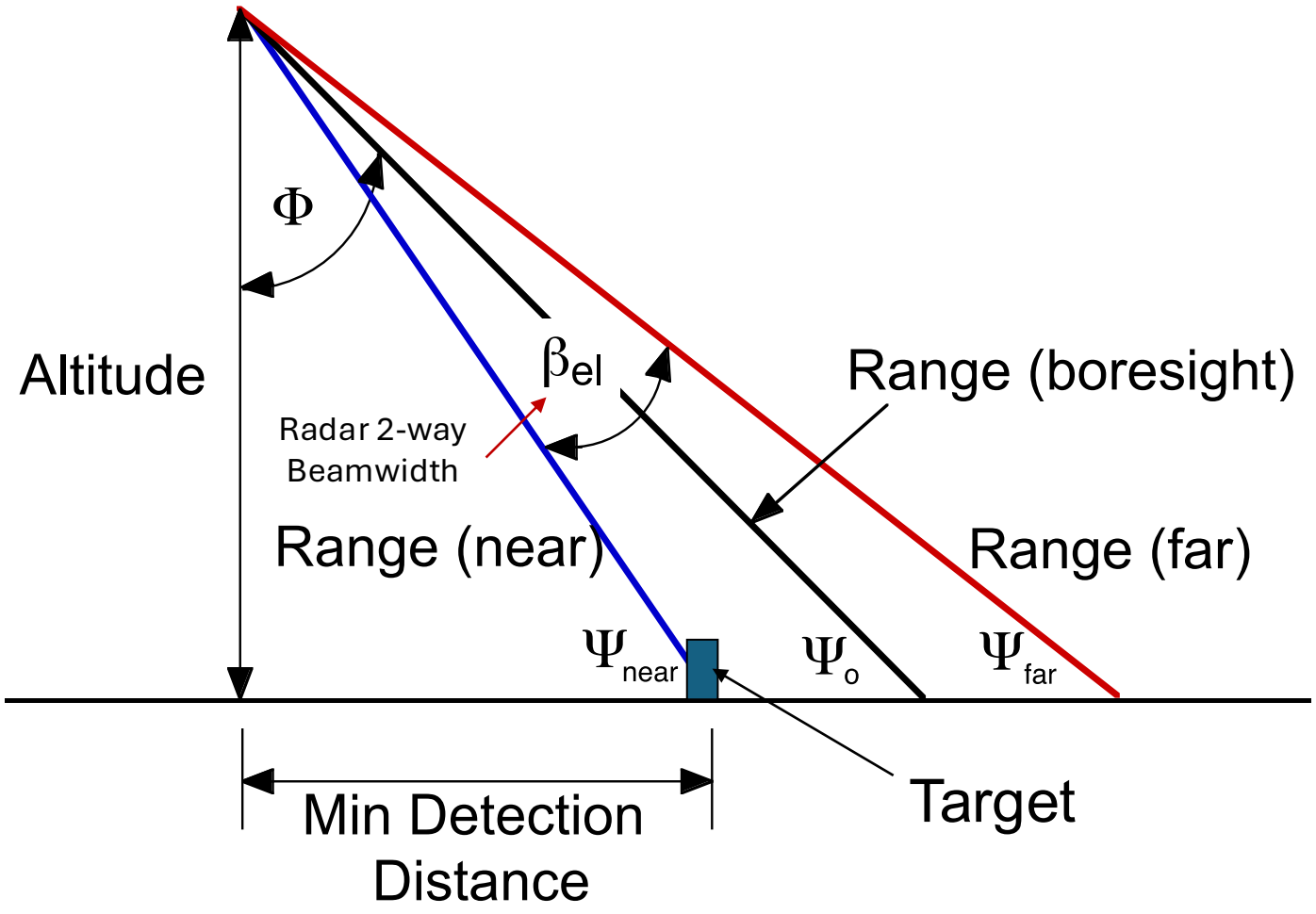


Antenna Footprint & Target Location for Worst Case (Min Warning Time)



Minimum Detection Distance Calc

Definition: Closest distance that a target may be before detection



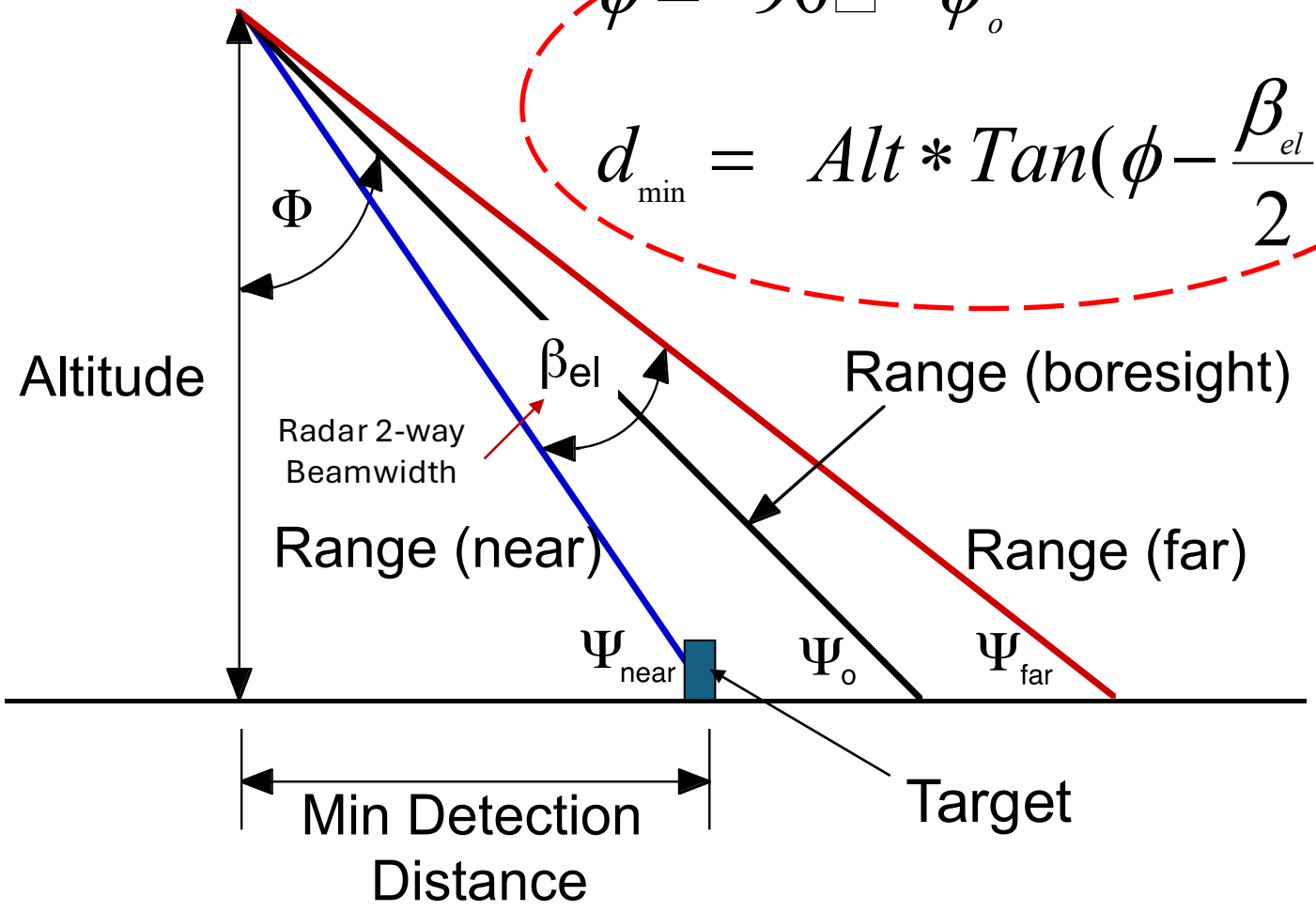
Minimum Detection Distance Calc

Definition: Closest distance that a target may be before detection

Boresight antenna pointing

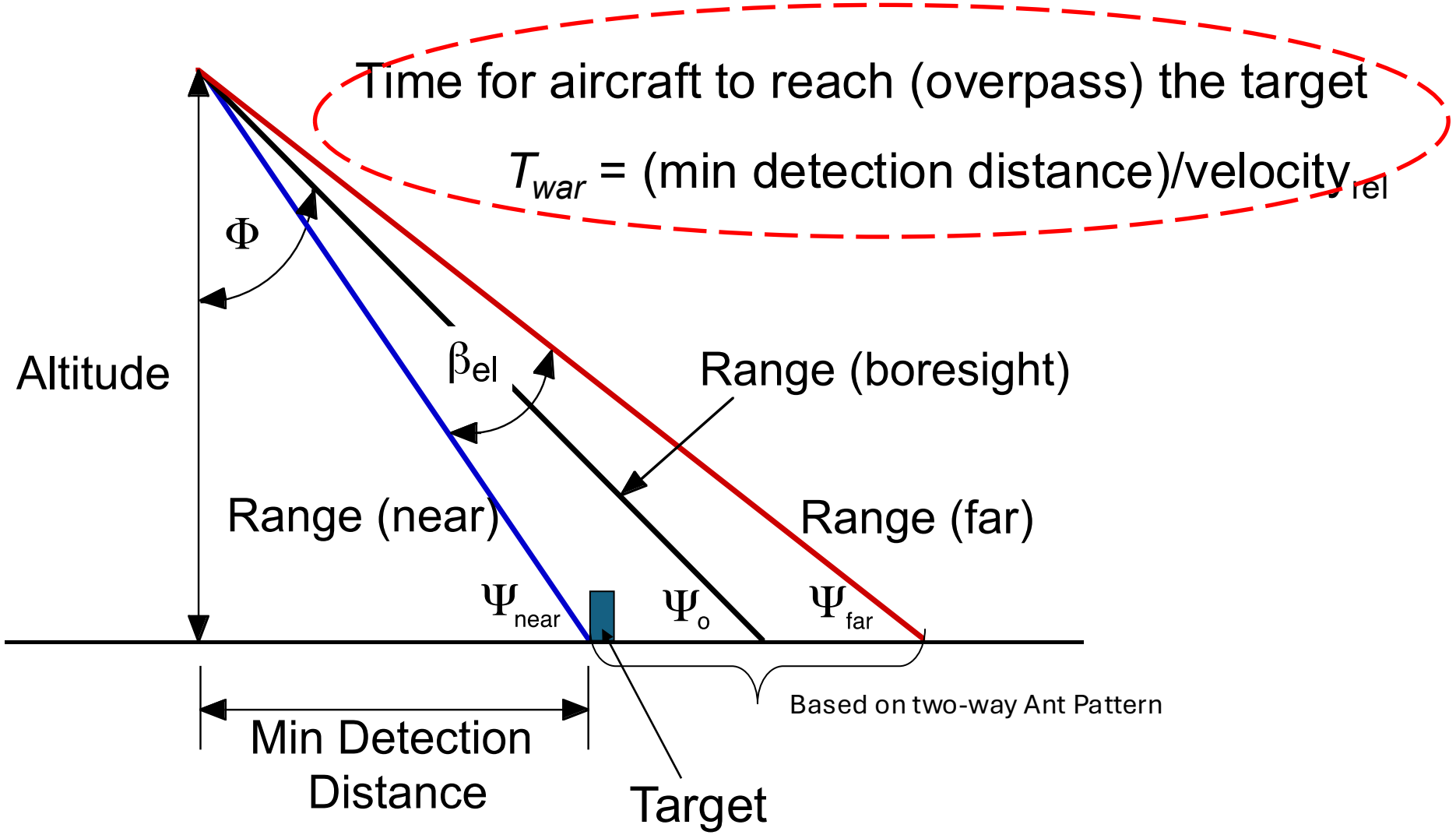
$$\phi = 90 - \psi_o$$

$$d_{min} = Alt * Tan(\phi - \frac{\beta_{el}}{2})$$



Minimum Warning Time (T_{war})

Definition: time delta between first target detection and the time of aircraft closest approach



Target Relative Velocity along UAV Track

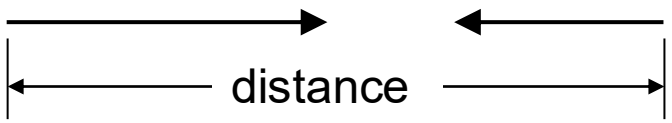
Calculate Relative Velocity

A = 10 m/s

(airplane)

B = 5 m/s

(target)



What is the relative velocity of B with respect to A?

10 m/s – (5 m/s*cos(180°)) = 15 m/s

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Calculate Relative Velocity

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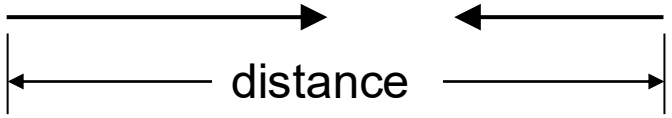

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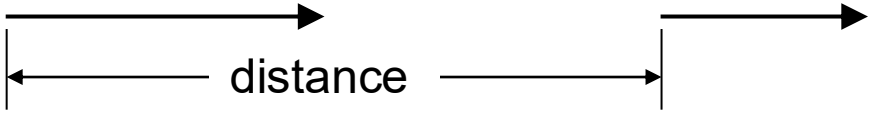
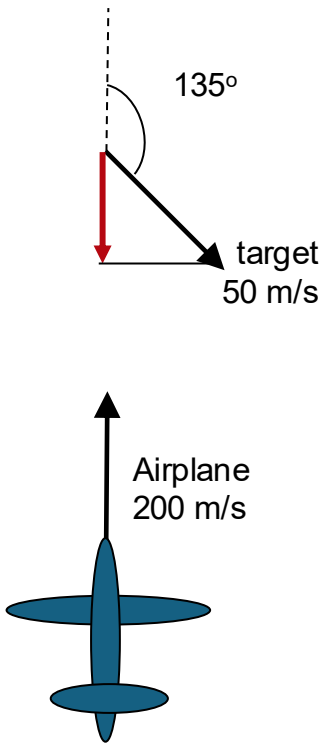


What is the relative velocity of B with respect to A?

10 m/s – (5 m/s*cos(0°)) = 5 m/s

Minimum Warning Time Calc cont.

Airplane flying at an altitude of 5000 meters with a velocity of 200 m/s due North, the grazing angle is 25°, Elevation HPBW is 10°. If a target is moving SSE (135° from North) at 50 m/s find the minimum warning time.

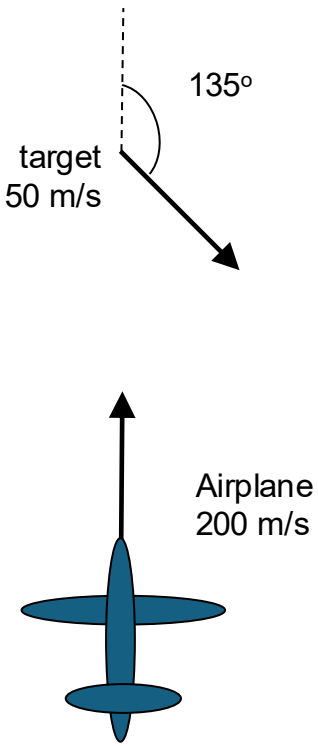


Calc relative velocity:

$$V_{rel} = 200m/s - 50m/s * \cos(135^o) = 235.4m/s$$

Minimum Warning Time Calc cont.

Airplane flying at an altitude of 5000 meters with a velocity of 200 m/s due North, the bore-grazing angle is 25°, Elevation HPBW is 10°. If a target is moving SSE (135° from North) at 50 m/s find the minimum warning time.

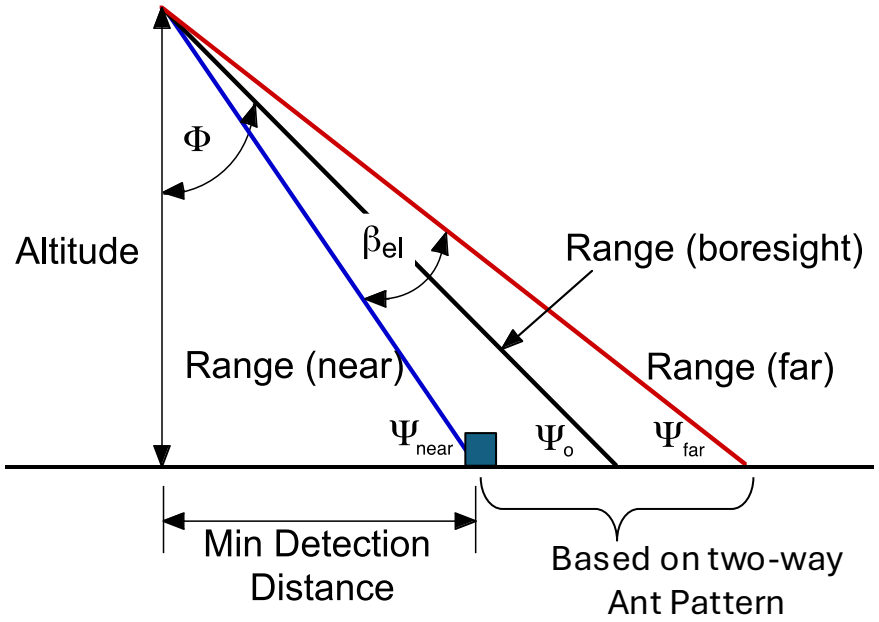


Calc. min distance (target @ R_{near}):

Cone ang @ Rnear =

$$\phi - \beta_{el}/2 = (90^o - 25^o) - \left(10^o/2\right) = 60^o$$

$$Dist_{\min-\det} = 5000m * \tan(60^o) = 8660.3 meters$$



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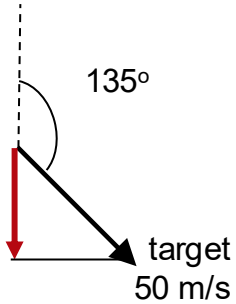
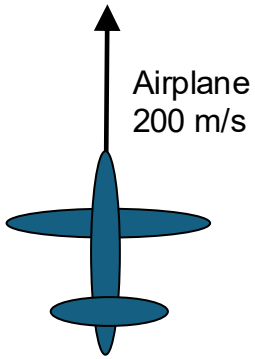


Calc relative velocity:

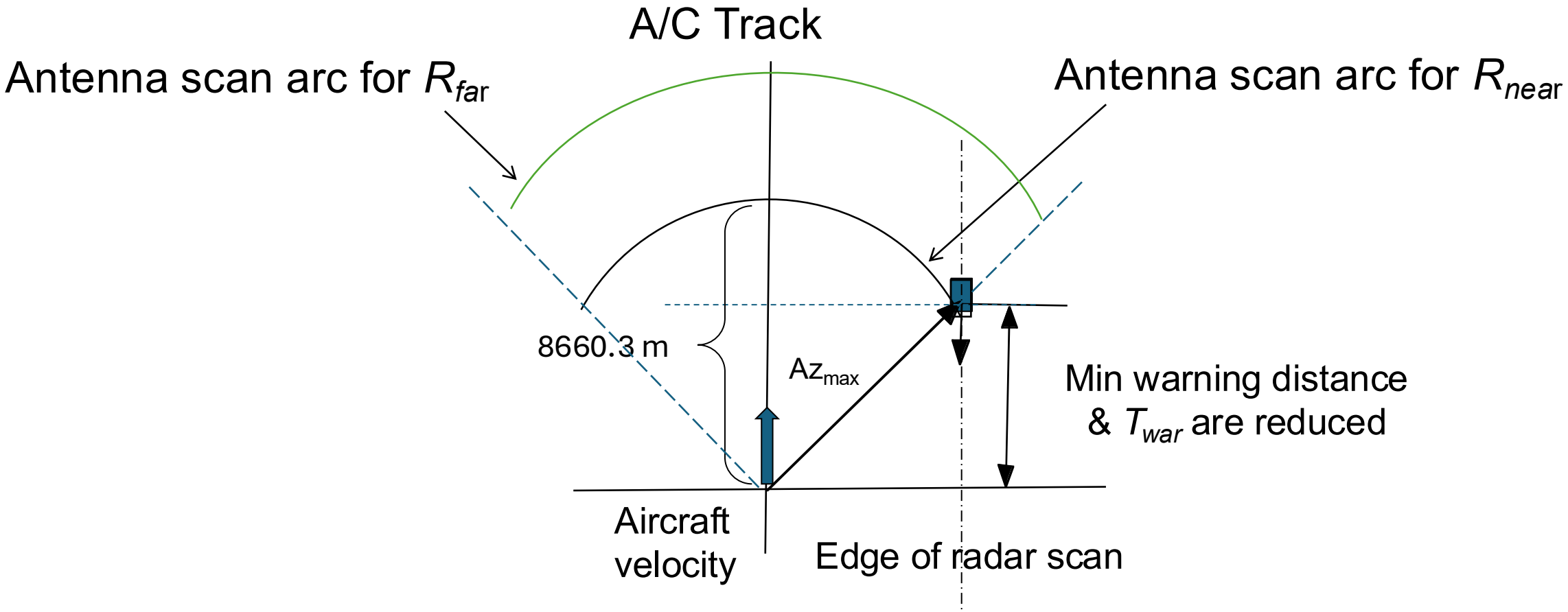
$$V_{rel} = 200m/s - 50m/s * \cos(135^o) = 235.4m/s$$

Calc min warning time:



$$t_{\min} = \frac{8660.3m}{235.4m/s} = 36.8\sec$$

Minimum Warning Time for Target Occurs @ Edge of Scan



$$T_{war_min} = T_{war_along\text{-}trk} * \cos(Az_{max})$$

Example-3 Minimum Spin Rate

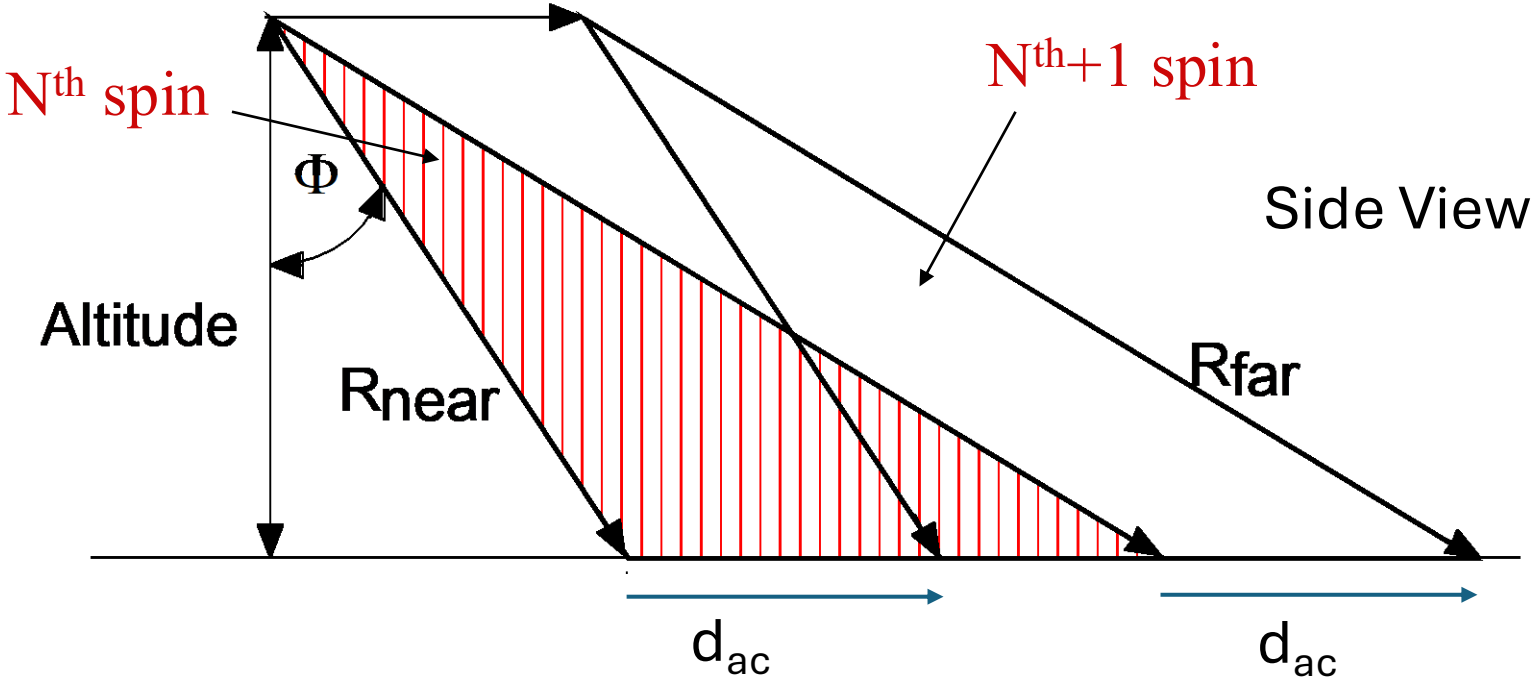
- After, you have selected an antenna boresight Ψ_o and the β_{EL}
  - Now for this antenna pointing configuration, calculate the minimum spin rate to ensure <u>not missing a target</u>

Minimum Antenna Footprint Overlap (for two consecutive scans) Required to Prevent Missing a moving Target

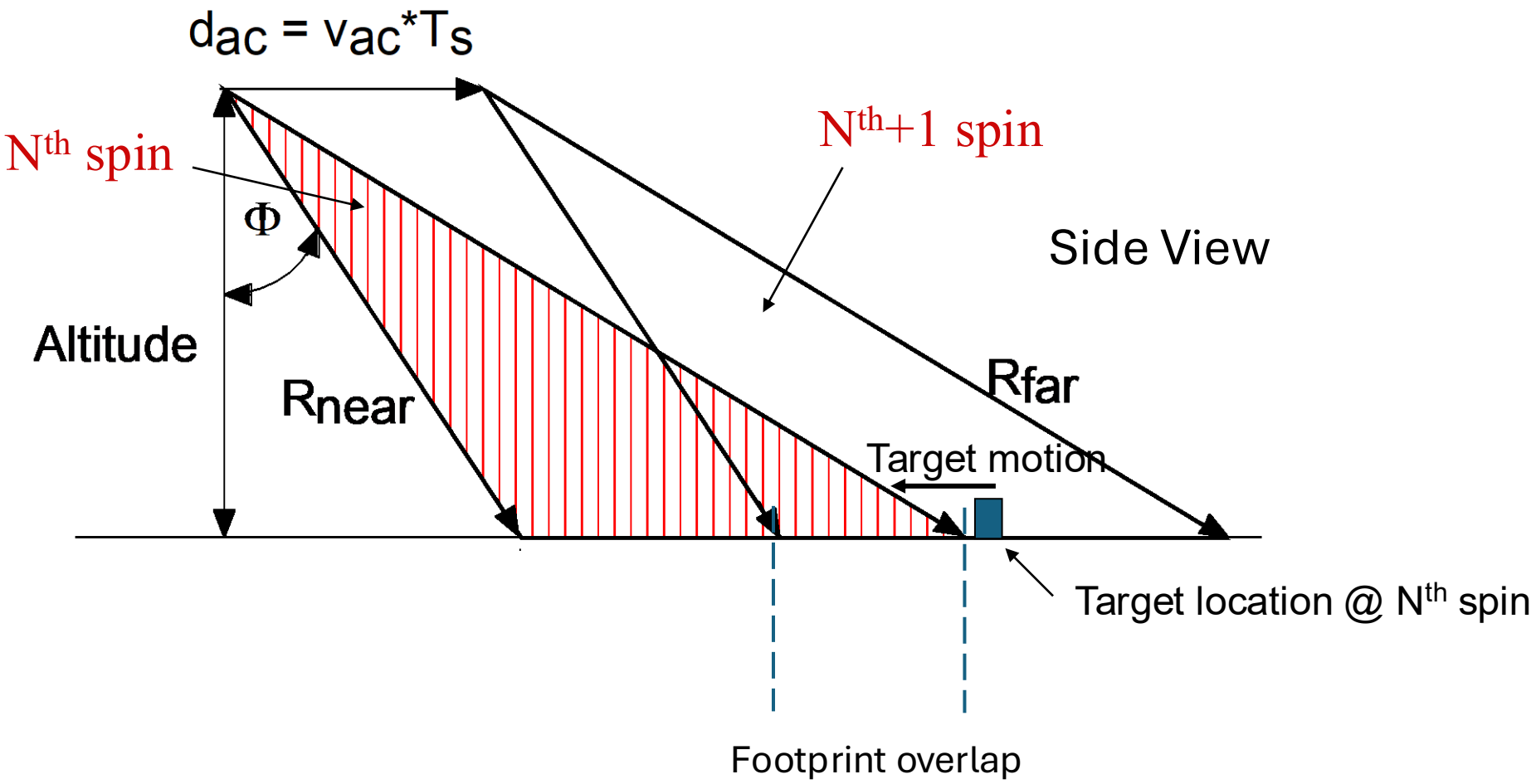
- Consider the spinning antenna footprints for 2 consecutive scans (spin #'s)

Distance that AC moves during one-antenna scan

$d_{ac} = v_{ac} * T_s$

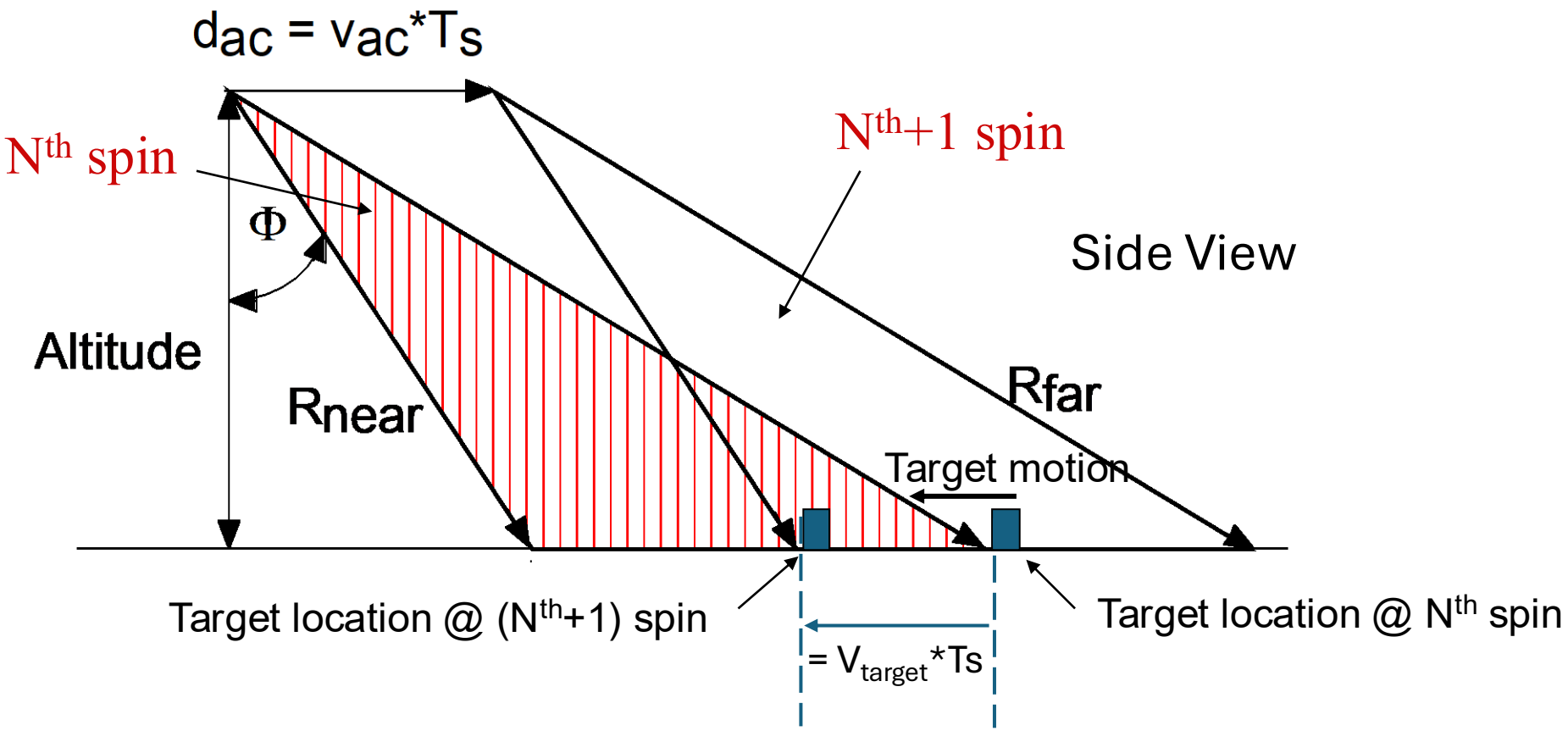


- Consider a target located "just outside" of ant footprint for the Nth spin
  - Ant must spin rapidly enough to ensure that a moving target will be observed
    - This requires that the footprints overlap between scans

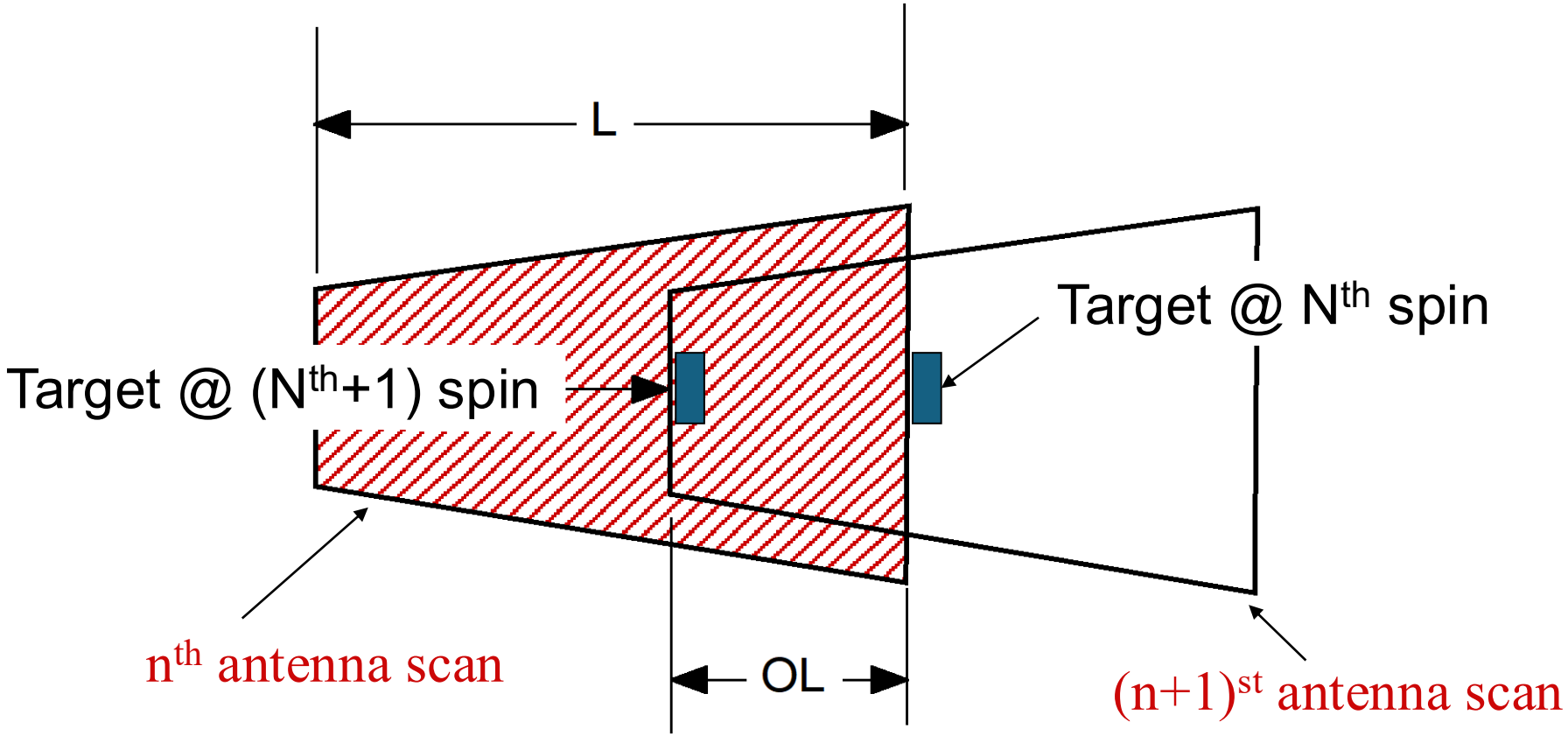


Minimum Antenna Footprint Overlap for two consecutive scans – to prevent missing target

- Target is "just within" the ant footprint for the (N^{th}+1) spin
  - if the antenna were spinning slower, the target would be missed!



Antenna Footprint Overlap for two consecutive scans - Top View



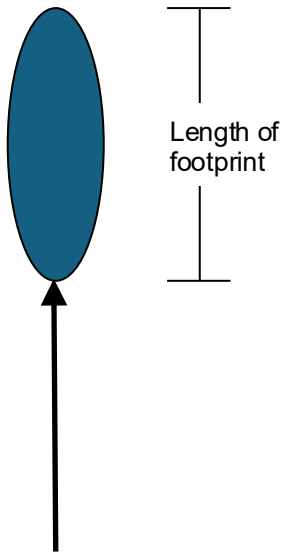
Calculate Antenna Footprint

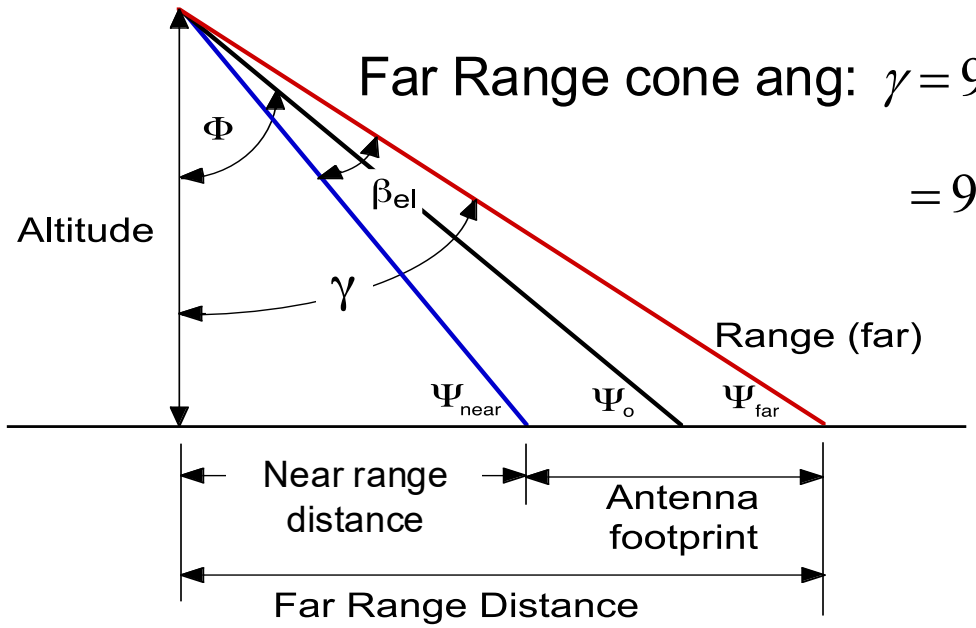
Boresight cone ang: $\phi = 90\square - \psi_o$

Far Range cone ang: $\gamma = 90\square - \psi_o + \left(\beta_{el}/2\right)$

$= 90\square - \psi_{far}$

Ant IFOV





Calculate length of footprint:

$$length_{footprint} = dist_{Range(far)} - dist_{Range(near)}$$

$$length_{footprint} = Alt * (\tan(\gamma) - \tan(\Phi - \beta_{el}/2))$$

Minimum Spin Rate without missing a target (i.e., just detecting the target once)

$$SpinRate(rpm) = \frac{60}{Period(\sec)}$$

$$Period(\sec) = \frac{length_{footprint}}{Velocity_{relative}}$$

$$V_{rel} = V_{airplane} - V_{target} * \cos(\theta)$$

where θ is the angle between the direction of flight of the airplane the direction of movement of the target

Worst case: θ = 180° (target & UAV approaching)

Minimum Spin Rate Req'd to detecting the target "n" times

$$SpinRate(rpm) = \frac{60}{Period(\sec)}$$

$$Period(\sec) = \frac{length_{footprint} / n}{Velocity_{relative}}$$

$$V_{rel} = V_{airplane} - V_{target} * \cos(\theta)$$

where θ is the angle between the direction of flight of the airplane the direction of movement of the target

Worst case: θ = 180° (target & UAV approaching)

Spin Rate to View Target for 3-scans by Varying Antenna El Dimension for Freq = 150 GHz

Case-A

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	1.50E+11	Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation	0.500	meter
Ant dia- azimuth		meter
Ant spin rate	80	rpm

Ant footprint length. M	302.24
footprint overlap scan to scan,	0.9173
Ant Spin Period, sec	0.78
Beta_El , deg (1-way pattern)	0.28

Spin Rate to View Target for 3-scans by Varying Antenna El Dimension for Freq = 150 GHz

Parameters not shown are constant

Case-A

Radar Design Parameters Choices		
Radar Freq	1.50E+11	Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation	0.500	meter
Ant dia- azimuth		meter
Ant spin rate	80	rpm
Antenna Azimuth +/- from fore-look		deg

Ant footprint length. M	302.24
footprint overlap scan to scan,	0.9173
Ant Spin Period, sec	0.78
Beta_El , deg (1-way pattern)	0.28

Case-B

Radar Design Parameters Choices		
Radar Freq	1.50E+11	Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation	0.200	meter
Ant dia- azimuth		meter
Ant spin rate	32	rpm

Ant footprint length. M	755.89
footprint overlap scan to scan,	0.9208
Ant Spin Period, sec	1.88
Beta_El , deg (1-way pattern)	0.69

Spin Rate to View Target for 3-scans by Varying Grazing Ang for Freq = 37 GHz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth				
Ant spin rate	15	50	100	115
Antenna Azimuth +/- from fore-look				

Optimizing # Scans that Target tracked by Varying Grazing Ang for Freq = 37 GHz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180
Design # scans target tracked	7	6	5	4

Increasing Spin Rate

Example-4 Time on Target (*TOT*)

Example Time on Target (*TOT*)

- Now that you have selected
  - An antenna boresight Ψ_o and the β_{EL} and the minimum spin rate to ensure <u>not missing a target</u>
  - What about the effect of *TOT*?
    - Increasing the *TOT* will maximize the pulses on target (*POT*), which will improve S/N (using non-coherent pulse integration)
      - $POT = TOT * PRF$

Constraints on PRF:

- Calculate the max PRF:
  - (assures only one pulse on the surface at a time - see Pulse Radar Geometry)

$$PRF_{max} = \frac{1}{(R_{far} - R_{near})/c + 2\tau}$$

Normally negligible

Constraints on PRF:

- Calculate the max PRF:
  - (assures only one pulse on the surface at a time - see Pulse Radar Geometry)

$$PRF_{max} = \frac{1}{\left(R_{far} - R_{near}\right)/c + 2\tau}$$

- Pulses on Target:

$$POT = PRF * \left[\frac{60}{spin-rate}\right] * \left[\frac{\beta_{az}}{360}\right]$$

Where spin-rate is revolutions/min and β_{az} is Ant Az HPBW, deg

Thus: spin rate, β_{el}, β_{az}, ψ_o - all interact and *POT* must be traded against other reqmts

Example-5 Prob of Detection (*Pd*)

Example P_d Trades

- Prob of detection (P_d) depends upon
  - $T/(C+N)$ & POT & P_{fa}

Example P_d Trades

- Prob of detection *(P_d)* depends upon
  - *T/(C+N) & POT & P_{fa}*
- Clutter cross section = *sig-o * pulse IFOV area*
  - Clutter sig-0 = *f(ψ)*
  - Pulse IFOV area (δR x δAz)
    - But $\delta Az = R * \beta_{az}$ and the target range = *f(ψ, Alt)*
    - and δR = *f(pulse width)*

Example P_d Trades

- Prob of detection *(P_d)* depends upon
  - *T/(C+N) & POT & P_{fa}*
- Clutter cross section = *sig-o * pulse IFOV area*
  - Clutter sig-0 = *$f(\psi)$*
  - Pulse IFOV area (δR x δAz)
    - But $\delta Az = R * \beta_{az}$ and the target range = *$f(\psi, Alt)$*
    - and δR = *f(pulse width)*
- Noise power depends upon the receiver bandwidth, which proportional to = 1/ (pulse width)

Example P_d Trades

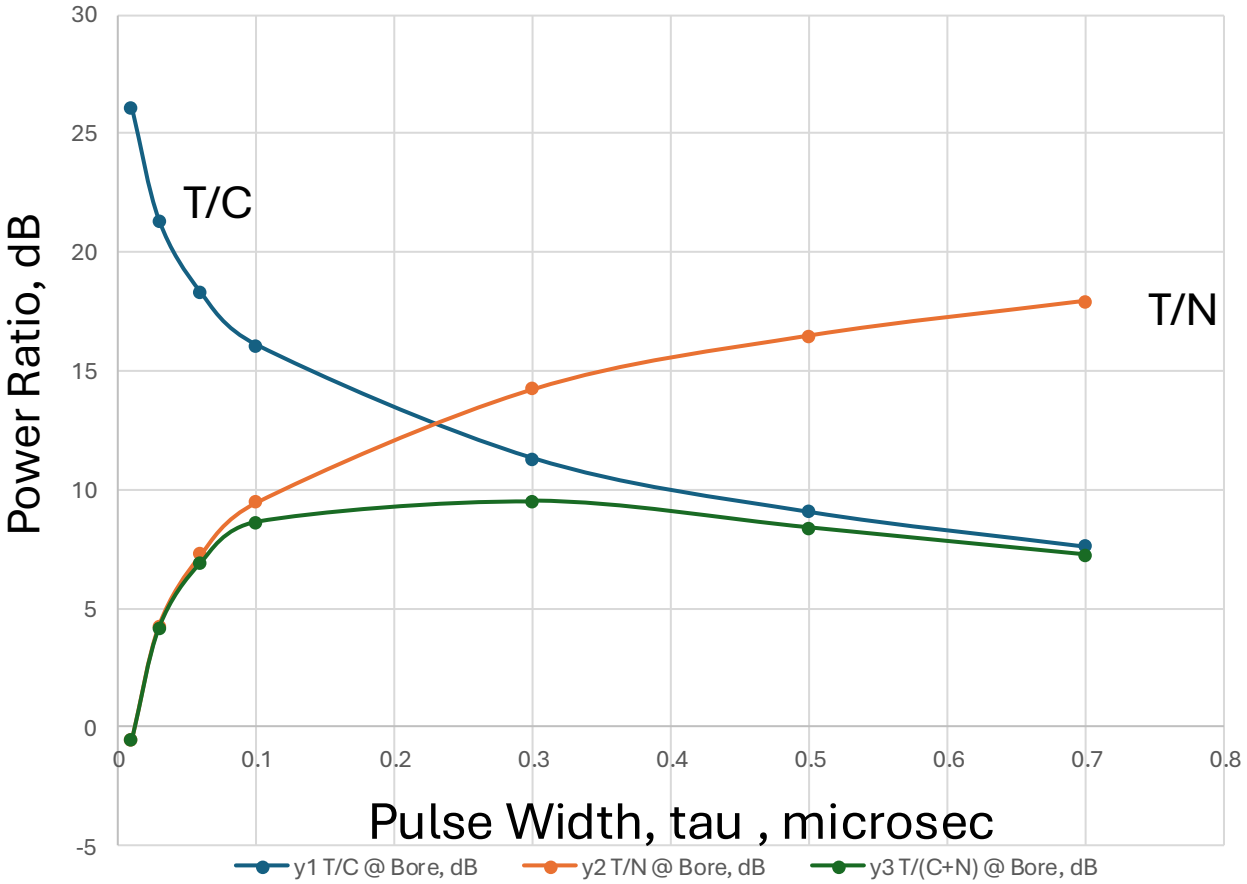
- Prob of detection *(P_d)* depends upon
  - *T/(C+N) & POT & P_{fa}*
- Clutter cross section = *sig-o * pulse IFOV area*
  - Clutter sig-0 = *f(ψ)*
  - Pulse IFOV area (δR x δAz)
    - But $\delta Az = R * b_{az}$ and the target range = *f(ψ, Alt)*
    - and δR = *f(pulse width)*
- Noise power depends upon the receiver bandwidth, which = 1/ (pulse width)
- Target cross section = *f(ψ)*

Signal to Noise Expectations: (with varying Pulse Width, tau)

All 7-cases are fully spec-compliant designs

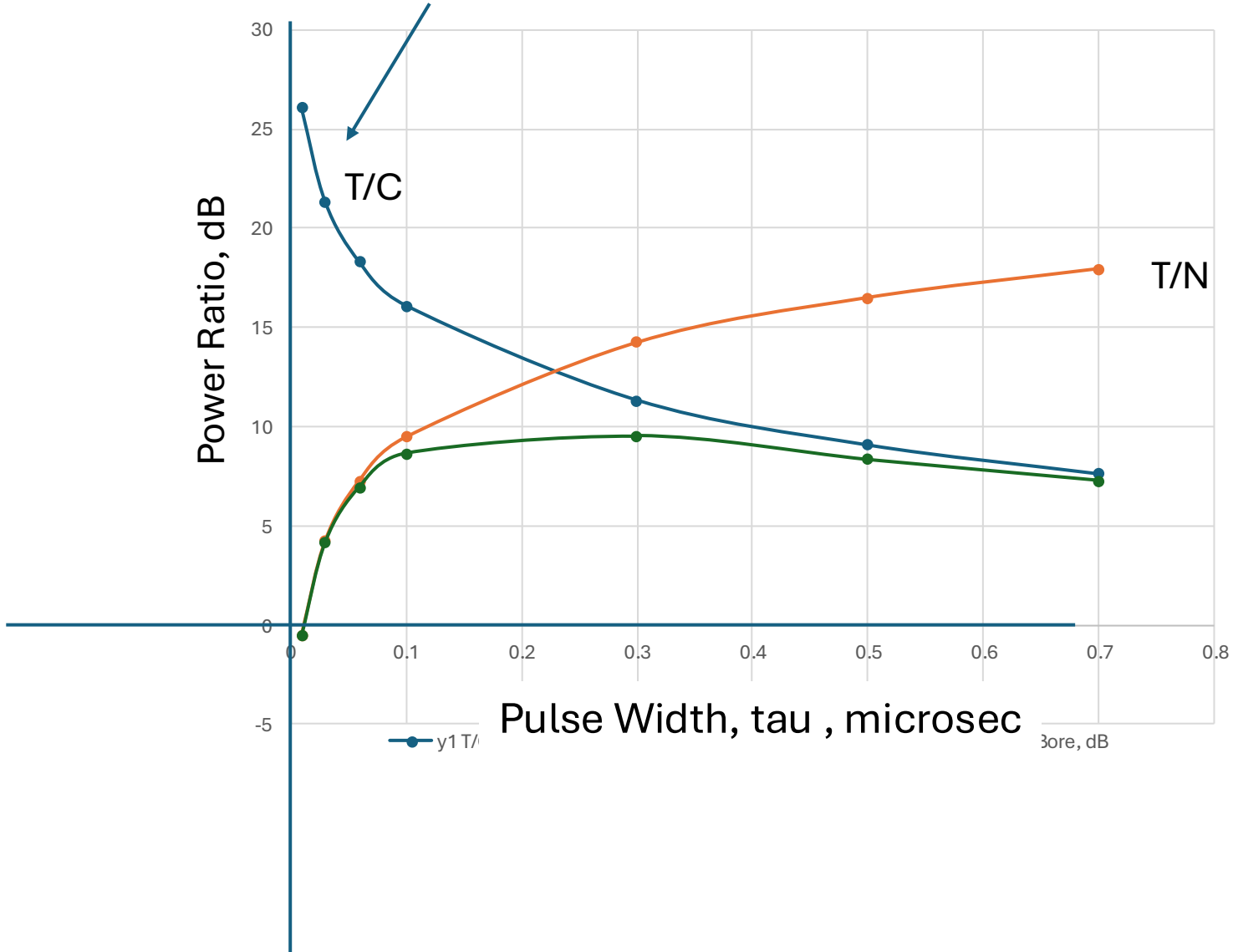
All Constant

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant_Bore_grazing_Ang		
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmt power		W
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer



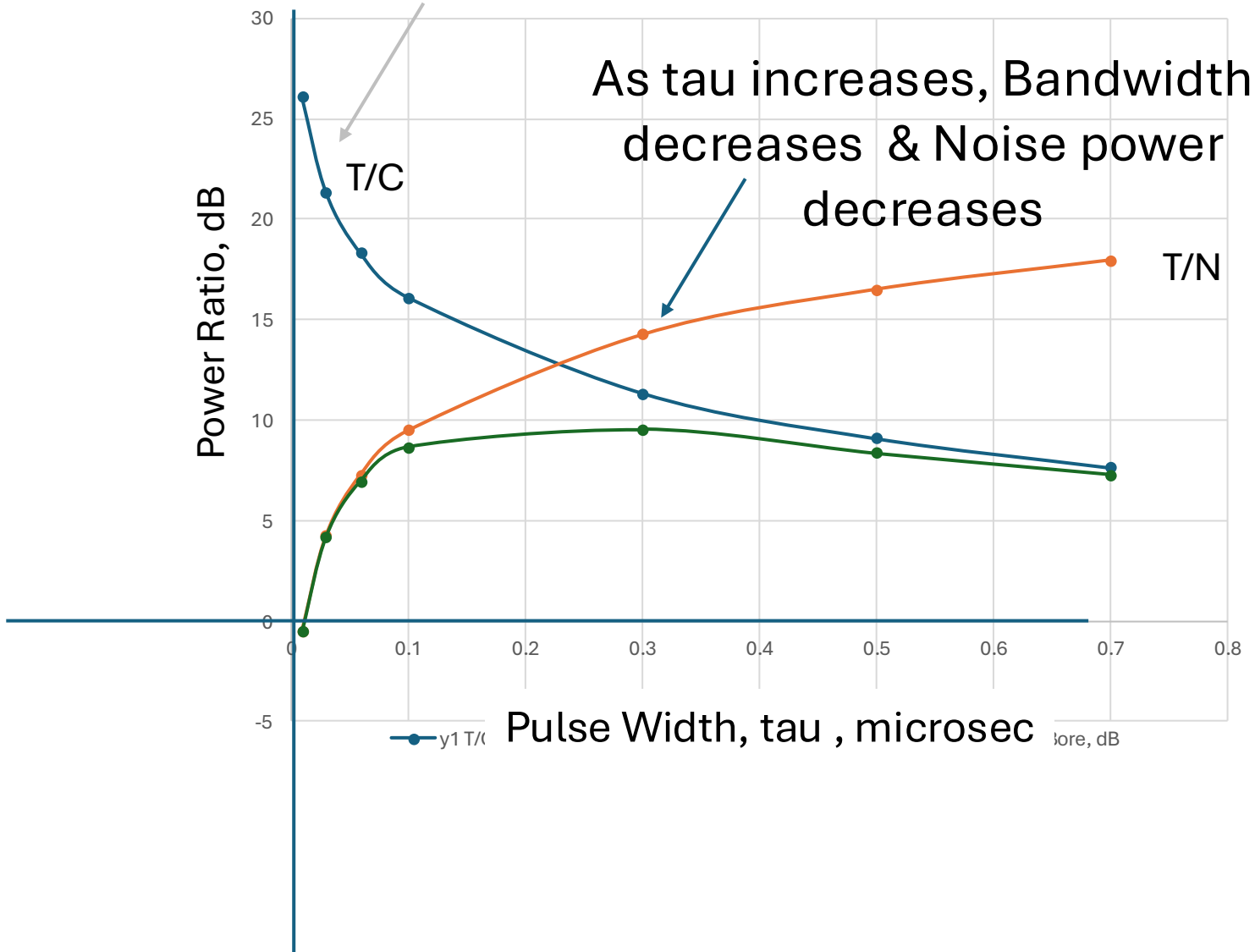
Signal to Noise Expectations: (with varying Pulse Width, tau)

As tau increases, pulse IFOV (δR) increases & Clutter increases

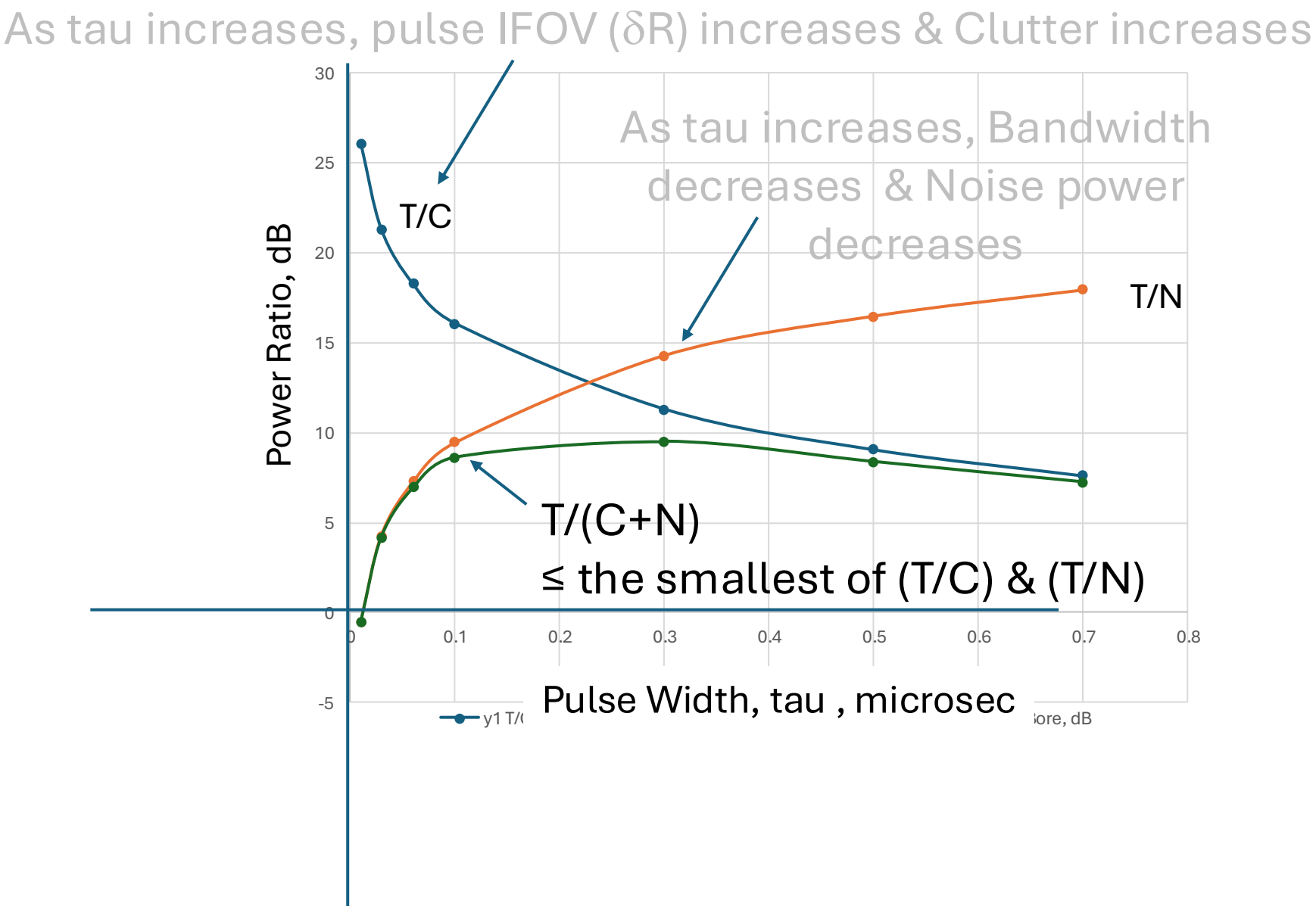


Signal to Noise Expectations (with varying Pulse Width, tau)

As tau increases, pulse IFOV (δR) increases & Clutter increases

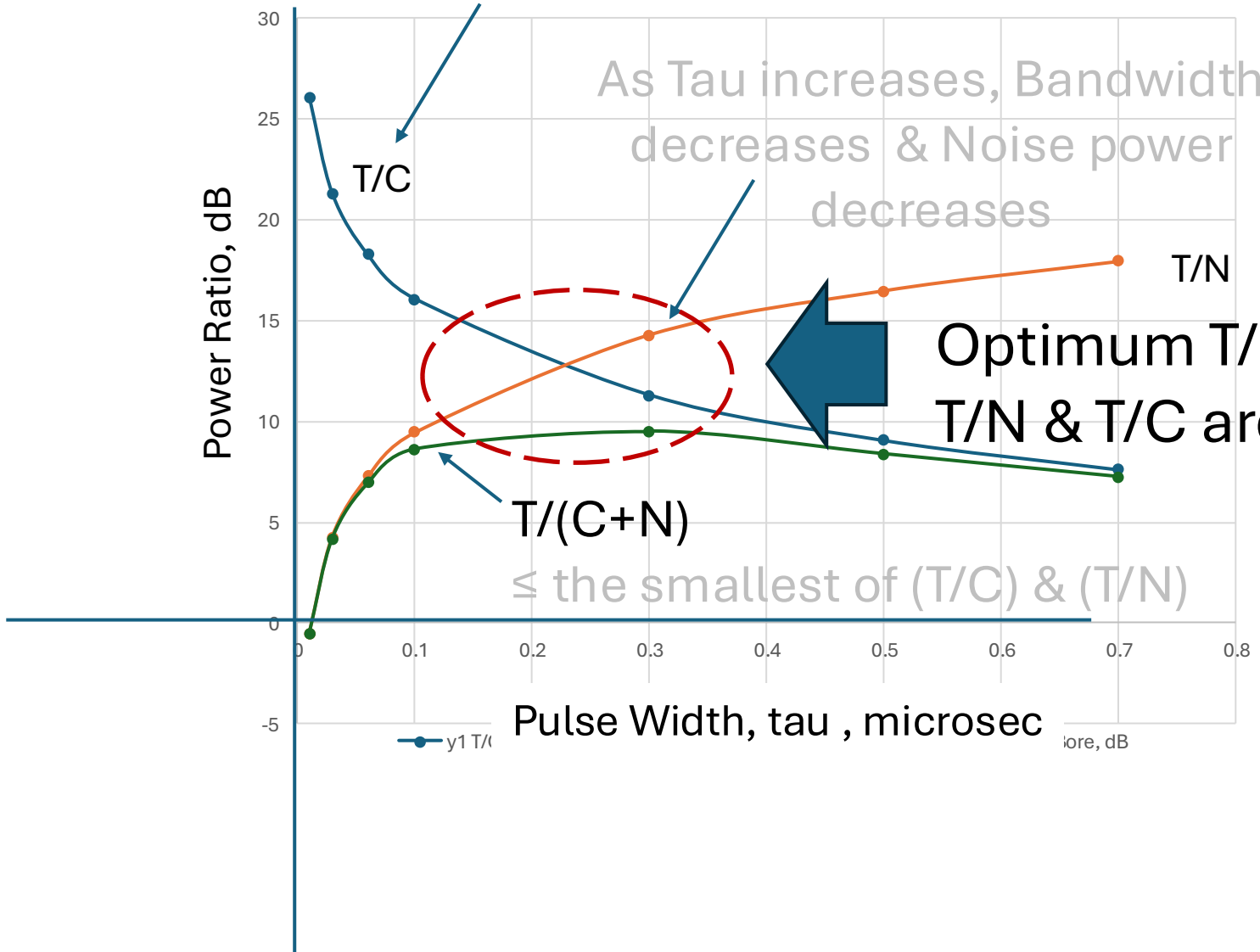


Signal to Noise Expectations (with varying Pulse Width, tau)



Signal to Noise Expectations (with varying Pulse Width, tau)

As Tau increases, pulse IFOV (δR) increases & Clutter increases

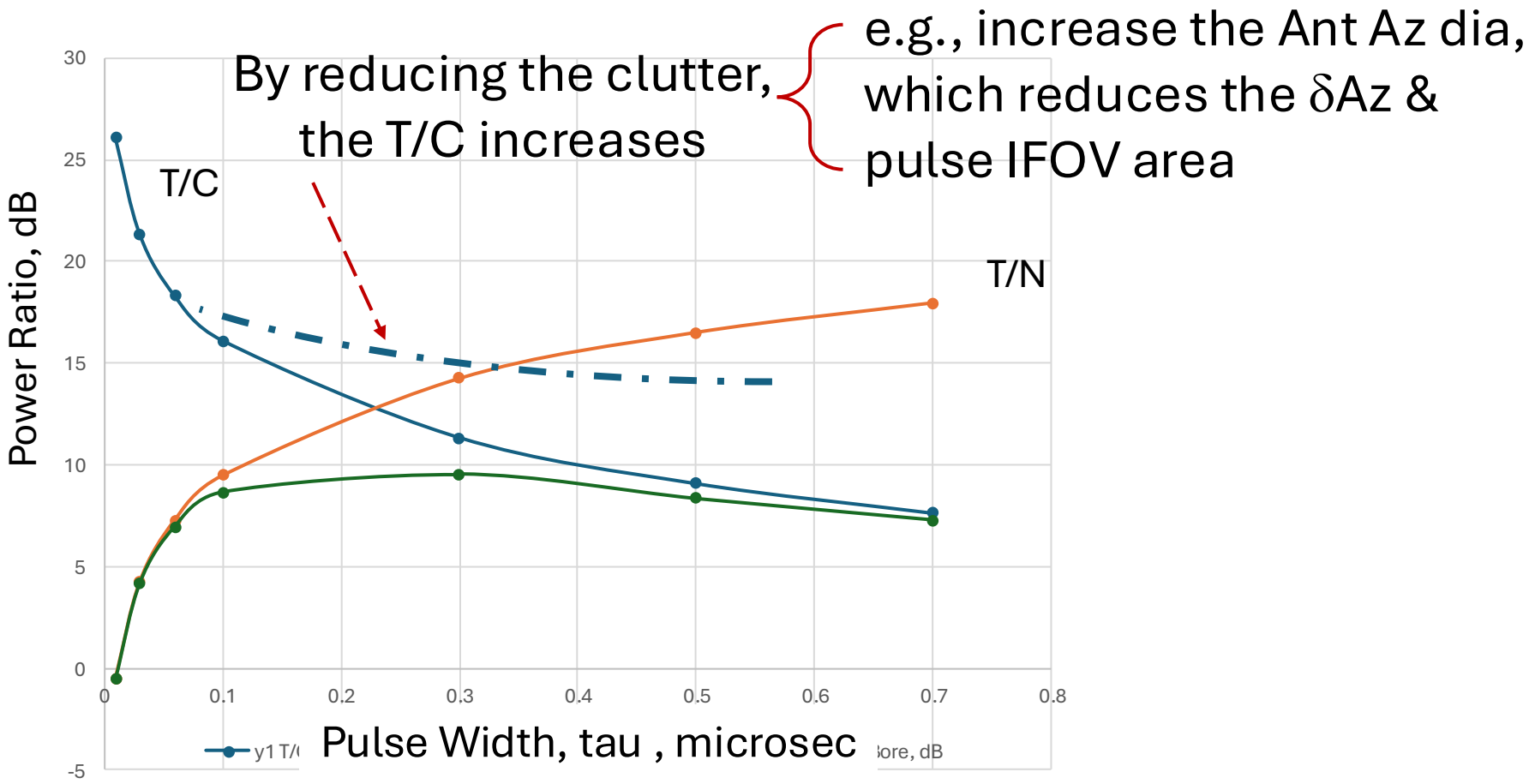


As Tau increases, Bandwidth decreases & Noise power decreases

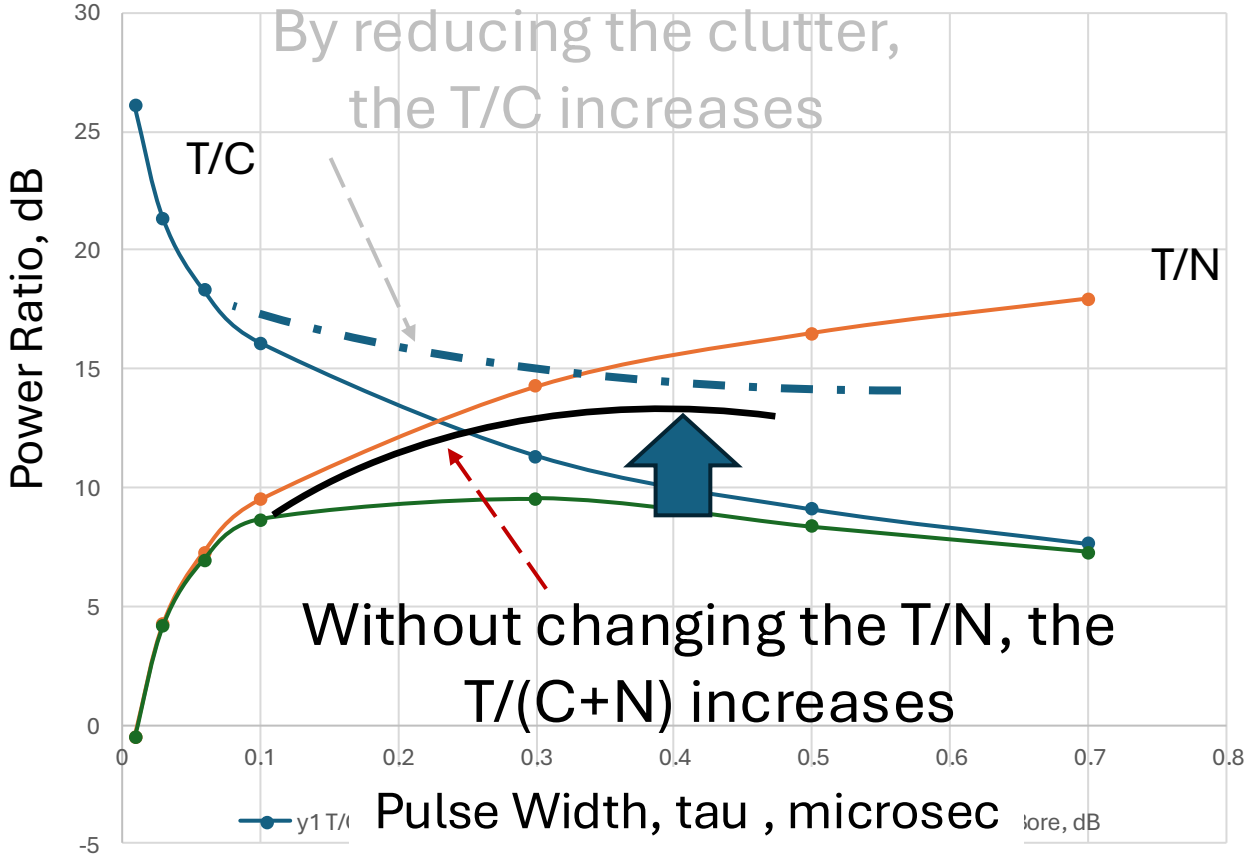
Optimum T/(C+N) occurs when T/N & T/C are within ~ 3 dB

≤ the smallest of (T/C) & (T/N)

Improvements in Signal to Noise



Improvements in Signal to Noise



Submit your test cases to validate your model before you conduct trade studies!